TEMPORALLY ORIENTED CONTENT GENERATION OF LEGAL DOCUMENTS VIA ARGUMENTATIVE ABSTRACTION

PHASE II REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Summarizing legal documents involves challenges like diverse argumentative structures and inter-document temporal dependencies. To find the right balance between semantic coherence, factual accuracy, and chronological fidelity while minimizing redundancy across the connected documents is not achievable with the current state of approaches. There is promise in the field of automatic text summarization, however, traditional extractive or abstractive approaches to summarization have challenges. A graph-based, argument-oriented framework is presented for summarizing from multiple legal documents that incorporates semantic analysis, argument mining, and reasoning with a knowledge graph. This framework constructs an argument-aware graph representation of legal documents, it uses Graph Neural Networks to encode discourse relations, and this representation takes into account the need to eliminate redundancy, select information, determine a temporal ordering, and verify significant physical facts in order to summarize the documents in a way that is coherent and legally faithful. It also successfully adapts to an Indian legal corpus. Experiments conducted on Multi-LexSum corpus demonstrate good performance in terms of ROUGE and BERTScore metrics, achieving a peak ROUGE-1 score of 45.14, while producing summaries that are concise, factually-coherent and temporally-ordered.

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CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

The legal field produces enormous amounts of written information, such as judgments, pleadings, transcripts, and legal memos [1]. Such documents can be lengthy, somewhat interrelated, and span a number of years, with extensive cross-referencing and complex chains of reasoning [30]. Manually reading and synthesizing all these materials is not only time-consuming, but there is also the possibility that an attorney, a researcher, or a student may overlook or fail to incorporate an essential fact [21]. The rapid growth of legal writing and the complexity of its reasoning template underscore the need for an automated function that can summarize these long, interrelated documents while aligning on facts [26].

1.2 CHALLENGES IN LEGAL MULTI DOCUMENT SUMMARIZATION

Recent research on automatic text summarization has examined both extractive and abstractive summarization methods using graph-based systems, neural architectures, and hybrid models. Graph-based methods such as TextRank [6] and EdgeSumm [39] aim to capitalize on structural relationships between sentences,

but do not capture deep semantic or argumentative relationships. Transformer-based models like BART [9] and Longformer [5] provide rich contextual representation, but are computationally expensive and require large annotated datasets. In law, domain-adaptive methods such as argument mining [22] and role-based extraction [17] have produced promising results in capturing discourse structures, yet face limitations in scalability and generalizability, especially in multi-document scenarios [27]. The challenge of multi-document summarization is particularly pronounced in the legal domain due to the need to preserve coherent chronology [12], argumentative integrity [20], and factual accuracy across related affidavits [26].

Both extractive and abstractive summarization approaches face significant challenges when applied to legal writing. Legal documents contain dense argumentation [1], temporal dependencies [12], cross-references [6], and specialized professional language that require domain-specific treatment. Extractive methods often fail to preserve the logical flow of legal reasoning [39], while abstractive systems may introduce unverifiable assertions or omit essential distinctions [9]. The repetitiveness of legal writing, such as restating arguments [30], citing the same precedents [21], or reiterating procedural directions [28], complicates the generation of summaries.

1.3 LIMITATIONS OF EXISTING APPROACHES

Legal documents also present additional challenges due to their argumentative justifications [12], strict factual accuracy requirements [20], and temporal event sequencing [22]. Current summarization systems often fail to effectively handle

redundancy [26], paraphrased material [6], or logical dependencies between related legal cases [39]. Moreover, Indian legal texts introduce regional language [21] and structural nuances [28], [19], adding further complexity. Overall, there remains a major gap in integrating argumentative abstraction [1], temporal structuring [12], and factual verification [9], [17] into a unified summarization framework.

Graph-based methods [6], [22] have inherent limitations in terms of semantic understanding which would lead to deep and informative abstraction and have higher likelihoods of errors when dealing with paraphrased data or semantically remote inputs. Converter based methods based on transformers, especially multi-granularity formatted transformers, have addressed some of these limitations. However, they are computationally expensive and have unpredictable performance in noisy environments. Hybrid approaches that leverage graphs have added even more complexity since they rely on external structured knowledge resources [9], which might be impractical in low-resource configurations. In the domain of law, while domain-specific models have shown encouraging results [5], [27], producing coherent summaries still relies primarily on fully annotated corpuses with limited cross-domain transferability. Also, even with some good notional approaches to defining and reasoning cross-domain matching, a formal representation of the knowledge is not yet possible [22], [17].

1.4 PROPOSED FRAMEWORK OVERVIEW

This paper proposes a new framework for temporally oriented content generation using argumentative abstraction. These limitations highlight several open research

questions that drive this work:

1. RQ1: How can argument roles (Claim, Evidence, Rule, Background) be incorporated into graph structures to improve legal summary coherence?
2. RQ2: Can GAT models be adapted to prioritize legally significant roles during embedding learning?
3. RQ3: How can redundancy detection be improved to handle paraphrased legal content across documents?
4. RQ4: What is the best way to combine graph-based and embedding-based importance scores across varied legal case types?
5. RQ5: How can evaluation metrics be designed to assess preservation of legal reasoning chains in summaries?

The proposed method constructs a graph-based representation, specifically a knowledge graph where argumentative roles such as claims, evidence, rules, and background information are represented as nodes. A Graph Attention Network (GAT)-based architecture [17] is then employed to capture relational and discourse-level structures, producing temporally ordered summaries that preserve both logical flow and factual alignment [27].

Our proposed approach addresses these existing methodological gaps by:

1. Merging argumentation mining with graph-based semantic representation.
2. Combining structural importance with graph-embedding methods.

3. Introducing explicit mechanisms to handle temporal ordering, and fact verification during summarization.

This work makes three key contributions to the legal knowledge extraction and summarization community.

1. It introduces an end-to-end pipeline that combines argument-aware graph construction, redundancy removal, and multi-criteria sentence selection to automatically generate legally coherent summaries of multi-document corpora.
2. It systematically integrates temporal information to ensure summaries accurately reflect the chronological order of legal events, a major shortcoming in general-purpose summarization systems.
3. Finally, to validate the approach, we evaluate it on the Multi-LexSum dataset of U.S. civil rights cases and demonstrate its adaptability to Indian legal contexts.

1.5 BACKGROUND

This domain presents unique challenges for multi-document summarization due to its reliance on argumentative structures, temporal sequencing, and strict factual accuracy. Hybrid methods like [12] have been proposed to balance factual coverage and interpretability. Neural architectures like [20] have shown promise in capturing long-range dependencies, though they remain limited in handling argumentative and temporal aspects. Studies focusing on Indian legal corpora

[21], [26] further reveal that region-specific judgments pose unique linguistic and structural challenges, making the summarization task even more complex. Another issue is redundancy in legal texts. Since judgments and related documents frequently restate arguments, cite precedents, and repeat procedural details, traditional summarization methods often result in repetitive content when it comes to summarizing multiple documents.

A system that generates concise, coherent, and factually reliable summaries can significantly enhance productivity and accessibility in the legal domain. Such summaries would support case preparation, academic study, and public understanding.

1.6 PROBLEM DEFINITION

This project introduces a specialized Multi-Document Summarization framework. It adopts an argument-oriented graph-based approach that identifies and extracts core legal arguments, maps their interconnections across documents, and aligns them in their correct chronological order. By integrating fact verification mechanisms, the system ensures that the generated summaries uphold integrity while highlighting essential reasoning and procedural flow. The resulting summaries are designed to provide legal professionals with a precise, coherent, and chronological synthesis of complex case material, enabling rapid and reliable navigation of extensive legal documents.

1.7 ORGANIZATION OF THE REPORT

The rest of the report is organized as follows: Section 2 will provide an extensive literature survey. Section 3 will provide a detailed description of methodology including preprocessing, argument mining, reasoning graph construction in two varying approaches with simple graphs as well as with Graph Attention Networks, redundancy removal, information selection, temporality ordering and fact checking. Section 4 provides experimental results and investigation of performance analysis for threshold performance optimization analysis with reasoning graph creation for redundancy filtering. Section 5 provides a discussion of social impact and sustainability. Finally, Section 6 summarizes contributions of our findings and recommendations for future work.

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORKS

Research in automatic summarization has explored diverse techniques, ranging from extractive graph-based methods to abstractive models enriched with external knowledge.

- **Graph-Based Foundations:** Graph methods have been widely used for summarization. El-Kassasa et al. [6] introduced *EdgeSumm* using TextRank and centrality, while Ragazzi et al. [22] applied graph clustering for redundancy reduction. Moro et al. [17] integrated sentence-level graphs with transformers, improving factual accuracy but often producing rigid summaries.

- **Transformer and Semantic Models:** To address semantic gaps, Zhang et al. [39] developed a multi-granularity extractor, and Huang et al. [9] combined knowledge graphs with BART for richer abstractive outputs. These improved contextual representation but required heavy computation and structured resources.

- **Legal Domain–Specific Adaptations:** Elaraby and Litman [5] proposed *ArgLegalSumm*, a dual-decoder transformer with argument mining, while Shen et al. [27] presented *Multi-LexSum*, focusing on legal discourse for

civil rights cases. Both showed domain strength but required large datasets or specific case structures.

- **Indian Legal Domain:** Shreyasi et al. [28] leveraged rhetorical roles for extraction, and Naik et al. [19] fine-tuned domain-specific transformers to capture contextual nuances. These highlight trends of weak supervision, discourse-aware methods, and domain-adapted transformers in Indian legal summarization.

As shown in Table 2.1, most prior works either focus on extractive summarization or rely heavily on external structured knowledge, leaving a clear gap in scalable, semantically grounded abstractive summarization for legal texts.

Recent studies have advanced graph-based summarization by incorporating Graph Neural Networks (GNNs) to explicitly learn relational dependencies through neural message passing.

- Yasunaga et al. [38] proposed Graph-based Neural Multi-Document Summarization, where documents are represented as sentence-level graphs and encoded using Graph Convolutional Networks (GCNs).
- Extending homogeneous sentence graphs, Wang et al. [35] introduced modeling documents as heterogeneous graphs comprising multiple node types such as words, sentences, and document-level nodes which demonstrates that heterogeneous GCN-based representations outperform sentence-only graph formulations.
- Zhao et al. [40] proposed a framework that constructs multi-granularity graphs, and applies attention-based message passing to dynamically

TABLE 2.1: Overview of Literature Survey

Paper	Method	Research Gap	Dataset	Limitation
El-Kassas et al. (2020) [6]	Graph-based summarization using TextRank and centrality-based features on edge-weighted graphs.	Absence of semantic graph representation and inability to perform abstractive summarization.	General (News)	Lacks semantic understanding and abstractive capability; purely extractive approach.
Huang et al., (2020) [9]	BART + Knowledge Graphs + semantic-driven Cloze-style RL reward.	Lack of methods for dynamic KG construction and integration in low-resource domains.	General (WikiHow, CNN-DM)	Relies on external structured knowledge graphs and complex reward modeling.
Elaraby & Litman, (2022) [5]	Argument mining + dual decoder Transformer. Joint training for summary and argument reconstruction.	Need for scalable unsupervised/semi-supervised argument mining in legal domain.	Legal (LEDGAR, BillSum)	Requires high-quality annotated datasets; limited generalization to unseen domains.
Elaraby & Litman, (2022) [5]	Graph-based sentence distillation + clustering across documents.	Gap in semantic alignment across documents and lacks abstraction.	General (News/Scientific)	Fails with paraphrased or semantically divergent content.
Moro et al., (2023) [17]	Sentence-level graph + abstractive Transformer decoder tuned for low-data domains.	Need for more flexible generation mechanisms in data-scarce environments.	Scientific/Medical (COVID datasets)	Performance heavily depends on extractive filtering; summaries are template-like.
Shen et al., (2022) [27]	Multi-granularity summary with factual + contextual + legal discourse analysis.	Lacks validation across broader legal domains with different structural patterns.	Legal (Civil Rights Lawsuits)	Only tested on civil rights lawsuits; limited generalization.
Shreyasi et al., (2025) [28]	Role-labelling of legal sections (facts, issues, arguments, judgment) + extractive summarization.	Only extractive; does not support abstractive summaries or multi-document reasoning.	Indian Supreme Court and High Court judgments (custom corpus).	Manual role schema definition required; scalability and adaptability to unseen cases remain weak.
Yasunaga et al., (2017) [38]	Graph-based MDS using sentence-level graphs and GCNs to capture global document structure and inter-sentence dependencies.	Lack of effective models to capture global document structure and inter-sentence dependencies in multi-document settings.	General (Multi-Document)	Homogeneous sentence-level graph structure (missing multi-level granularity).
Wang et al., (2020) [35]	Extractive summarization using Heterogeneous Graph Neural Networks.	Need to extend homogeneous sentence graphs to multi-level, heterogeneous graphs for richer relational modeling.	General	GCNs aggregate neighborhood information uniformly, potentially limiting salience selection.
Zhao et al., (2022) [40]	Extractive summarization using Multi-granularity Heterogeneous Graph Attention Networks (GATs) for dynamic, attention-based message passing.	Need for selective, attention-based weighting of inter-node relationships to enhance content selection over uniform GCN aggregation.	General	Increased model complexity and computational cost due to attention mechanism.
Bastings et al., (2017) [2]	GCNs used as encoders to integrate syntactic dependency trees (structured linguistic knowledge) into Neural Machine Translation.	Lack of effective methods for leveraging GCNs to encode structured linguistic knowledge for improved sequence-to-sequence tasks.	MT (Syntactic)	Applied to Machine Translation, requiring adaptation for summarization; limited to explicit syntactic relations.

emphasize informative connections to achieve superior performance over traditional GCN-based approaches.

- Bastings et al. [2] recommended syntactic dependency trees are modeled as graphs and integrated into neural encoders via graph convolution. Although applied to machine translation, this work establishes a foundational paradigm for leveraging GCNs to encode structured dependencies.

Collectively, these neural graph-based approaches demonstrate the effectiveness of GCN- and GAT-based architectures in modeling relational structure, hierarchical dependencies, and global context. This gap motivates the integration of GCN and GAT frameworks to capture the complex relational and discourse structures inherent in legal documents.

Table 2.2 provides a comparative overview of methodologies employed in selected research works related to legal and multi-document summarization. The columns represent key methodological dimensions.

TABLE 2.2: Comparison of methodologies used in selected research works related to legal and multi-document summarization

Paper	Abstractive Summary	Graph-based Representation	Argument Mining	Multi-document	Deep Learning / Neural Models
El-Kassas et al., (2020) [6]	-	Yes	-	Yes	-
Huang et al., (2020) [9]	Yes	-	-	Yes	Yes
Elaraby & Litman, (2022) [5]	Yes	-	Yes	-	Yes
Ragazzi et al., (2022) [22]	Yes	Yes	-	Yes	Yes
Moro et al., (2023) [17]	Yes	Yes	-	-	Yes
Shen et al., (2022) [27]	-	-	-	Yes	Yes
Shreyasi et al., (2025) [28]	-	-	Yes	-	Yes
Yasunaga et al., (2017) [38]	**-**	**Yes**	**-**	**Yes**	**Yes**
Wang et al., (2020) [35]	**-**	**Yes**	**-**	**Yes**	**Yes**
Zhao et al., (2022) [40]	**-**	**Yes**	**-**	**-**	**Yes**
Bastings et al., (2017) [2]	**-**	**Yes**	**-**	**-**	**Yes**

2.2 RESEARCH GAPS

While there has been progress on multi-document summarization, the current strategies demonstrate some significant weaknesses.

1. **Limitations of Graph-based Approaches:** Graph-based approaches [6], [22] usually have specific semantic exposure or lack the deeper knowledge needed to support an abstract or robust summarization process, and also tend to be limited by either paraphrased or more distant interference.

2. **Challenges with Transformer Models:** The increasing popularity of multi granularity transformer based models has helped overcome many challenges, but with too much of an associated computational and run time burden, plus, performance against noisy methods is seriously unpredictable. And where knowledge graph-based approaches [9] have been employed, they create larger selection challenges as they become dependent on some other complex external structured resource whilst working in low resource areas.

3. **Constraints of Legal Domain:** Legal domain–focused approaches show some promise [5], [27], but generating longer, well-structured summaries still depends heavily on large-scale annotated datasets, with limited prospects for transferability across contexts. Moreover, a robust and flexible cross-domain matching framework has yet to be precisely established or clearly defined, and current problem areas remain difficult to manage despite offering significant potential for earlier solutions [22], [17].

This incomplete picture suggests a demand for a robust and flexible summarization framework, capable of withstanding noisy and low resource domains, with low stifling costs, combining semantic understanding with computational efficiency, and willingly employing anything as a source.

2.3 RESEARCH OBJECTIVES

This project aims to address the identified research gaps by developing a robust, multi-faceted summarization framework. Our primary objective is to create a

system that overcomes the limitations of current approaches by integrating **argumentative abstraction** with a **temporal-oriented content generation** process

- **Mitigate the limitations of existing graph-based methods** by using a **Graph Neural Network** to create a richer semantic graph representation. This allows the model to prioritize key argumentative roles, like claims and evidence during the embedding process, moving beyond simple similarity to capture the true structure of legal reasoning.
- **Enhance the efficiency and effectiveness of multi-document summarization** in the legal domain by combining graph-based importance scores with embedding-based representativeness scores. This hybrid approach ensures that the generated summaries are not only structurally sound but also semantically comprehensive, avoiding the high computational costs and data dependencies of purely transformer-based models.
- **Tackle redundancy and improve coherence** by implementing a refined redundancy elimination algorithm that uses embedding similarity to detect and remove paraphrased or semantically similar content across documents.
- **Ensure factual accuracy and narrative integrity** by incorporating a temporal ordering module that arranges the selected summary sentences chronologically and a fact verification mechanism that validates key claims against source documents. This guarantees that the final output is both reliable and easy to follow, making legal knowledge more accessible and actionable.

2.4 FEASIBILITY STUDY

1. **Data Resources:** The proposed legal document summarization system is feasible due to the availability of quality datasets and domain-specific resources. Benchmark datasets such as Multi-LexSum provide large-scale and structured legal documents. Indian law judgments and case briefs can be scraped from reliable sources like the Supreme Court of India website https://www.sci.gov.in/case-status-case-no/, ensuring access to domain-relevant data for fine-tuning and evaluation.

2. **Technical Requirements:** The system can be efficiently executed on GPU-enabled environments such as Google Colab, or dedicated servers, which provide the computational power required to process large datasets and complex models like transformers and graph neural networks. All core modules of the pipeline can be implemented using well-established Python libraries such as Hugging Face Transformers, PyTorch Geometric, NetworkX, and scikit-learn.

3. **Availability of Pre-trained Models:** The project benefits from the availability of pre-trained transformer models (e.g., PEGASUS, LED, BART), which can be fine-tuned for domain-specific tasks. This reduces the need for training from scratch.

The combination of accessible data, scalable infrastructure, robust libraries, and pretrained models makes the proposed system both technically and practically viable.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed system is designed to produce accurate, concise, and context-aware summaries of legal documents through a combination of semantic analysis, argument mining, and knowledge graph-based reasoning, as illustrated in Fig. 3.1.

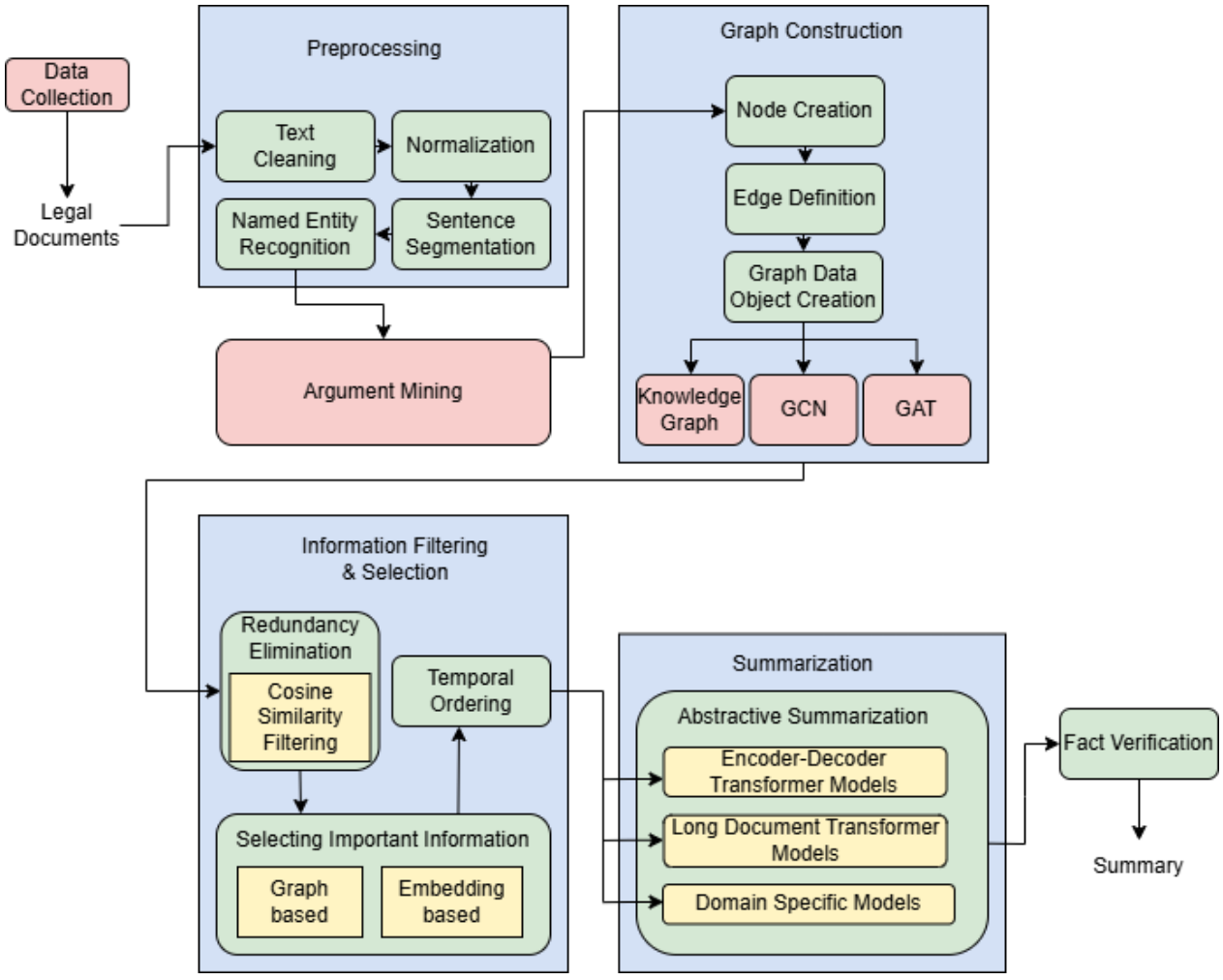

FIGURE 3.1: System architecture illustrating the overall summarization pipeline.

The process begins with preprocessing, where the input legal text is cleaned and segmented into sentences and clauses. Argument mining is then applied to detect and classify argumentative units such as claims, premises, evidence, and counterarguments, ensuring logical reasoning. These extracted elements are mapped into a graph-based representation. An abstractive summarization model then generates coherent summaries by leveraging both the original text and the enriched graph. The final output retains key facts, arguments, and chronological flow while eliminating redundancy, providing a clear, reliable summary.

Algorithm 1 summarizes the entire workflow framework formally. The initial step of the pipeline is data preprocessing to clean up and split and structure an input corpus into a set of sentences. Next, an argument mining module processes each sentence to assign semantic roles (such as claim, evidence, rule and background) to facilitate a discourse-aware representation of the input sentences. Following this, a matching graph of the sentences on a sentence basis is created. Then, the matching graph is encoded with Graph Neural Networks (GCN/GAT) to create an encoded representation for capturing contextual and relational dependencies in the document. The encoded representations are used to determine a hybrid importance score by combining structural centrality and semantic relevance, followed by redundancy filtering to remove redundant content. To ensure the procedural narrative is preserved, the highest-ranked sentences are selected from the list and ordered by their time stamps to sequentially order them to maintain the procedural narrative. An abstractive transformer performs final summarization on the highest-ranked sentences to create a coherent summary of the selected sentences. Finally, after the summary has been generated, a fact verification step is performed to verify that the information contained in the summary is supported by the respective evidence.

Input: Legal document corpus T, similarity threshold θ, redundancy threshold τ, summary length k

Output: Final summary S_{final}

1: Preprocess document T to obtain cleaned sentence set $S = \{s_1, s_2, ..., s_n\}$

2: Perform argument mining to assign role labels $r_i = \text{RoleClassifier}(s_i), \quad R = \{r_1, r_2, ..., r_n\}$

3: Construct graph $G = (V, E)$ where each node represents a sentence $v_i \leftrightarrow s_i$, and edges are formed using similarity $E = \{(v_i, v_j) \mid \cos(e_i, e_j) > \theta\}$

4: Encode graph using GNN (GCN/GAT) to obtain sentence representations $\quad H = \{h_1, h_2, ..., h_n\} = \text{GNN}(G, E)$

5: Rank sentences using combined graph importance and semantic relevance scores $\quad score(v_i) = \alpha \cdot score_g(v_i) + (1 - \alpha) \cdot score_e(v_i)$

6: Apply redundancy filtering using threshold τ to obtain candidate set $R_s = \{s_i \in S \mid \max_{s_j \in R_s} \cos(e_i, e_j) < \tau\}$

7: Select top-k sentences and arrange them using temporal ordering to obtain $\quad S_{sel} = \text{Order}\big(\text{TopK}(R_s, score, k)\big)$

8: Generate abstractive summary S_{abs} using transformer-based model $S_{abs} = f_{\text{trans}}(S_{sel})$

9: Perform fact verification on S_{abs} to ensure factual consistency $S_{final} = \text{Verify}(S_{abs}, E), \quad \text{label}(c_i) \in \{\text{Supported, Refuted, NEI}\}$

10: **return** S_{final}

Algorithm 1: Argument-Aware Legal Document Summarization Pipeline

3.1 DATA PREPROCESSING

Preprocessing is the foundational stage in the pipeline, ensuring that the raw legal text is cleaned and structured for further processing. Preprocessing begins by splitting the raw legal text corpus T into individual sentences to ensure manageable processing units. Each sentence is then converted to lowercase to maintain consistency. Punctuation marks, special symbols, and stopwords are removed to eliminate noise and retain only meaningful words. After cleaning, stemming or lemmatization is applied to reduce words to their base forms, allowing semantically similar words (e.g., argue, argued, arguing) to be treated uniformly. Entity recognition is also performed, as in Mentzingen et al. [16], to identify entities relevant to the case. Libraries such as *spaCy* and *NLTK* are used for tokenization, entity recognition, and stopword removal. This produces a set of cleaned sentences $S = \{s_1, s_2, \ldots, s_n\}$, ready for embedding generation, argument extraction, and graph construction.

3.2 ARGUMENT MINING

Argument mining is the process of automatically identifying and extracting argumentative components from textual data. Each cleaned sentence $s_i \in S$ is passed through a trained classifier that computes a prediction score indicating its likelihood of belonging to a particular argumentative role. Based on this score, the sentence is labeled as one of $\{$*Claim*, *Evidence*, *Rule*, *Background*$\}$. An example output showing sentence classification results with raw and final labels

(e.g., Claim, Evidence, Rule, Background) along with confidence scores is shown in Fig. 3.2

```
✅ Saved 139 labeled sentences to classified_sentences.json
Sentence: XVI LISTED On:31/10/2023 :12 :46 CIVIL APPELLATE JURISDICTION PETITION FOR SPECI...
 → Raw: Evidence (0.64), Final: Background

Sentence: 12230 of 2023 (With Prayer For Interim Relief) VASHIST NARAYAN KUMAR ....
 → Raw: Rule (0.79), Final: Rule

Sentence: Petitioner Versus THE STATE OF BIHAR ....
 → Raw: Claim (0.98), Final: Claim

Sentence: Respondents REPORT The matter above mentioned was listed before the Hon'ble Cour...
 → Raw: Evidence (0.96), Final: Evidence

Sentence: Issue notice, returnable within four weeks....
 → Raw: Rule (0.95), Final: Rule
```

FIGURE 3.2: Illustration of argument mining to identify claims, evidence, rules, and background.

If the classifier's confidence falls below a threshold τ, the sentence is assigned the default label *Background*. This sentence-level role labeling forms the basis for reconstructing the argument structure in legal documents, enabling downstream tasks such as structured summarization and argument graph construction. Recent studies [7], [37], [29] show how segmenting legal text into argumentative zones improves summary relevance.

The Sentence Role Labeling Algorithm (Algorithm 2) assigns argumentative roles to each sentence, including Claim, Evidence, Rule, and Background. This classification allows the downstream components to reorganize argumentative structure within the legal document. The threshold parameter τ controls whether predictions with lower confidence will be classified as background.

Input: Cleaned sentences S

Output: Labeled sentences with roles (Claim, Evidence, Rule, Background)

foreach sentence $s_i \in S$ **do**
  Compute prediction score using classifier.
  Assign role $r_i \in \{\text{Claim, Evidence, Rule, Background}\}$.
  if confidence $< \tau$ **then**
    Assign $r_i = \text{Background}$.
  end
end

Algorithm 2: Sentence Role Labeling Algorithm

3.3 KNOWLEDGE GRAPH CONSTRUCTION

Knowledge graph construction creates a structured, graph-based representation of entities, their attributes, and relationships. Formally, a knowledge graph is defined as $G = (V, E, L)$, where V is the set of nodes (entities), $E \subseteq V \times V$ is the set of edges (relationships), and L is a labeling function assigning semantic types or values to nodes and edges. An sample knowledge graph generated after argument mining is shown in Fig. 3.3

In the legal domain, knowledge graphs represent parties, statutes, events, and rulings with temporal annotations to capture case progression. Such graphs enable complex reasoning, support fact verification, and serve as the backbone for graph-based summarization [10]. They also facilitate explainability by making the connections between legal arguments transparent.

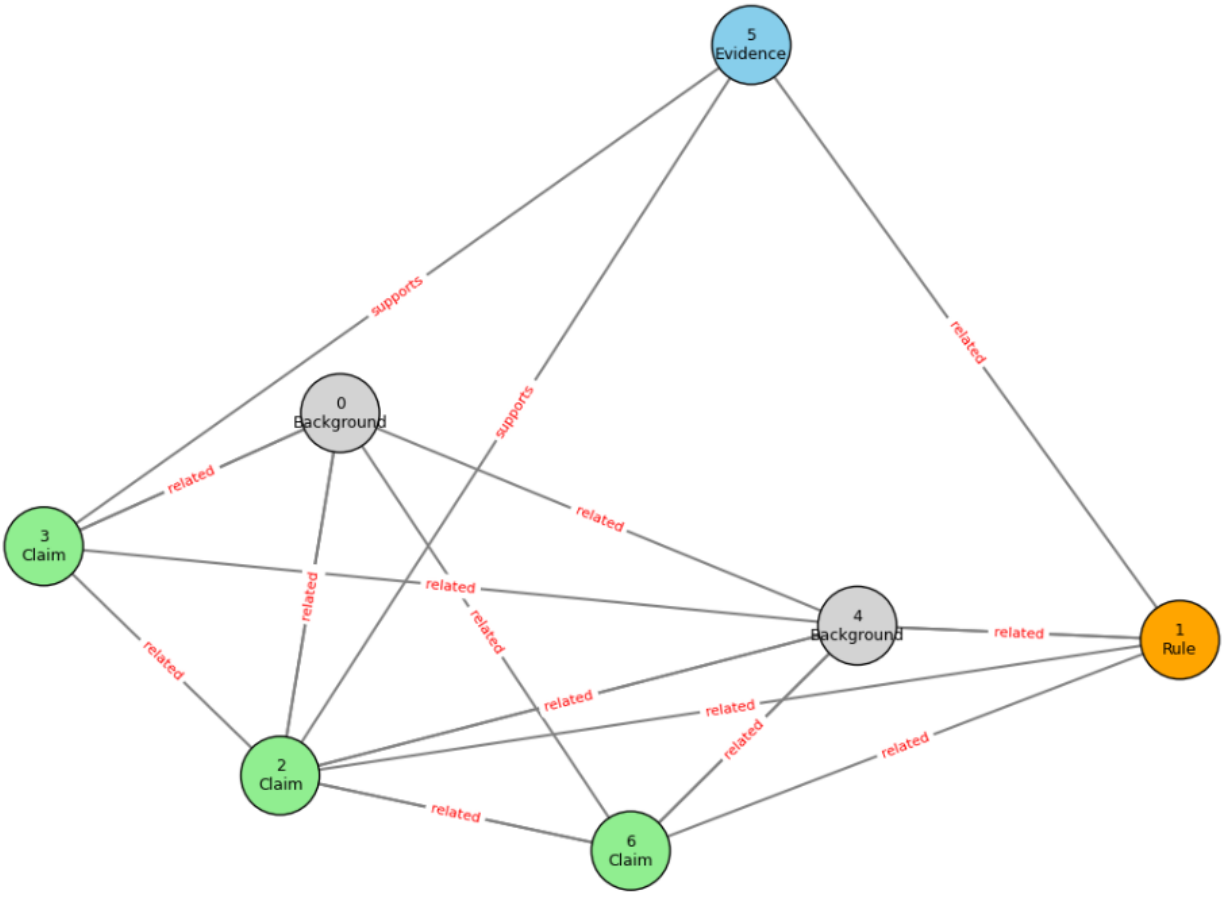

FIGURE 3.3: Knowledge graph generated after argument mining.

Algorithm 3 supplies the labeled input for the knowledge graph building. The Simple Knowledge Graph Algorithm produces a graph in which the nodes represent sentences and edges define semantic similarity. The similarity threshold θ is used to define whether or not two sentences will be connected, therefore, it directly impacts the density and the clustering behaviour of the graph.

Input: Labeled sentences with embeddings

Output: Graph $G = (V, E)$

Initialize G.

foreach sentence s_i **do**
  Add node v_i with attributes {text, role, embedding}.

end

foreach pair (s_i, s_j) **do**
  Compute similarity $sim(v_i, v_j)$.
  if $sim(v_i, v_j) > \theta$ **then**
    Add edge e_{ij} with weight $sim(v_i, v_j)$.
  end

end

Algorithm 3: Simple Knowledge Graph Algorithm

In the simple graph-based method, the labeled sentences are first used to construct a graph $G = (V, E)$. The library *NetworkX* is used for graph creation, and the Sentence Transformer *all-MiniLM-L6-v2* is used for embeddings. Each sentence s_i corresponds to a node v_i with attributes {text, role, embedding}. Similarities between node embeddings are computed, and edges are created when similarity exceeds a threshold θ. This results in a sentence similarity graph where edges capture semantic closeness, used to derive the important information and summary.

In the legal domain, knowledge graphs represent parties, statutes, events, and rulings with temporal annotations to capture case progression. Such graphs enable complex reasoning, support fact verification, and serve as the backbone for graph-based summarization [10]. They also facilitate explainability by making the connections between legal arguments transparent. Hierarchical frameworks

that employ knowledge graphs for the retrieval of case law documents are capable of modelling the complex relationship between case law documents and supporting structured reasoning across multiple case law documents [32].

3.4 GRAPH NEURAL NETWORKS (GNNs)

Graph Neural Networks (GNNs) [36] represent a class of deep learning models designed to operate directly on graph-structured data. In the context of natural language processing and document summarization, GNNs are utilized to explicitly model the complex relationships between text units, such as sentences or concepts, by treating the document as a graph. The goal of the GNN is to propagate and aggregate feature information across this graph to derive a robust, context-aware representation for each sentence, which can then be used by a classifier to predict whether the sentence should be included in the final summary.

As illustrated in Fig. 3.4, the constructed sentence-level graph is enriched with argumentative role labels and semantic embeddings, and processed through GNN layers to learn context-aware representations. These representations are subsequently used to compute sentence salience scores for selecting the most informative and legally relevant content for summary generation.

3.4.1 Graph Convolutional Networks (GCNs)

The Graph Convolutional Network (GCN) [11] is a GNN architecture that generalizes the convolution operation from image processing to graphs. A GCN

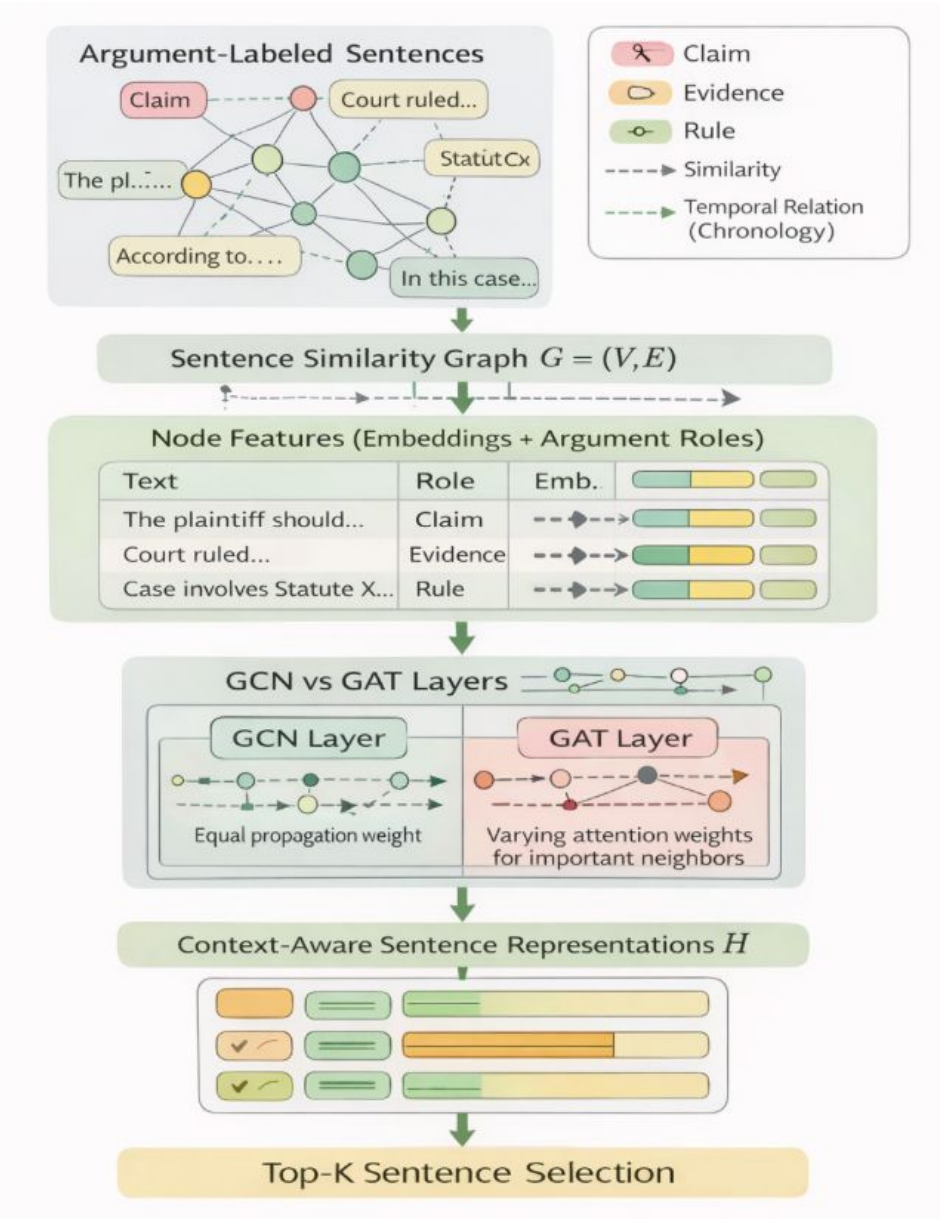

FIGURE 3.4: Argument-Aware GNN Architecture for Legal Summarization

layer learns a representation for a node by aggregating the feature information from its neighbors and its own previous state. Algorithm 4 specifies the matrix X contains the initial feature vectors of the sentences, and $\tilde{A}$ is the adjacency matrix with self-loops, normalized to $\hat{A}$ for propagation. At each layer, information from neighbouring sentences is combined using $\hat{A}$ and transformed using weights $W^{(l)}$ to obtain updated representations $H^{(l)}$. The final representations $H^{(L)}$ are then

used to compute sentence importance scores through a sigmoid function.

Input: Sentence Feature Matrix $X \in \mathbb{R}^{N \times F}$, Adjacency Matrix $\tilde{A}$ (with self-loops), Number of layers L, Weights $W^{(l)}$

Output: Final Sentence Embeddings $H^{(L)}$, Sentence Ranking Scores

$\tilde{D} \leftarrow$ Diagonal degree matrix of $\tilde{A}$.

$\hat{A} \leftarrow \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$

$H^{(0)} \leftarrow X$

for $l = 1$ **to** L **do**

  $Z^{(l)} \leftarrow \hat{A}H^{(l-1)}W^{(l)} \quad H^{(l)} \leftarrow \mathrm{ReLU}(Z^{(l)})$

end

$H_{final} \leftarrow H^{(L)}$

$S_{scores} \leftarrow \mathrm{Sigmoid}(H_{final} \cdot W_{rank})$

return S_{scores}

Algorithm 4: Graph Convolutional Network (GCN) for Sentence Ranking

3.4.2 Graph Attention Networks (GATs)

Graph Attention Networks (GATs) [33] enhance the approach by introducing a self-attention mechanism, allowing the model to assign different weights (importance scores) to different neighbors during the aggregation step. This mechanism addresses a major limitation of GCNs, which treat all neighbors equally (or based only on degree normalization). Algorithm 5 specifies the matrix X contains the sentence features and $\tilde{A}$ defines the connections between

sentences. In each layer, the features are first transformed using $W^{(l)}$, and attention scores are computed between each sentence and its neighbors using the vector $\vec{a}^{(l)}$. These scores are normalized to obtain weights α_{ij}, which are used to combine neighboring features and update the representation $H^{(l)}$. The final embeddings $H^{(L)}$ are then passed through a sigmoid function to produce sentence ranking scores.

Input: Sentence Feature Matrix $X \in \mathbb{R}^{N \times F}$, Adjacency Matrix $\tilde{A}$,
Number of layers L, Weights $W^{(l)}$, Attention Vector $\vec{a}^{(l)}$

Output: Final Sentence Embeddings $H^{(L)}$, Sentence Ranking Scores

for $l = 1$ **to** L **do**

  $H'^{(l-1)} \leftarrow H^{(l-1)} W^{(l)}$

  foreach neighbor j of node i **do**

    $e_{ij} \leftarrow \text{LeakyReLU}(\vec{a}^{(l)^T} [H'_i \| H'_j])$

  end

  Normalize Attention Scores α_{ij} (Attention Matrix $A^{(l)}$)

  foreach neighbor j of node i **do**

    $\alpha_{ij} \leftarrow \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})}$

  end

  $H_i^{(l)} \leftarrow \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} H'_j\right)$

end

$H_{final} \leftarrow H^{(L)}$

$S_{scores} \leftarrow \text{Sigmoid}(H_{final} \cdot W_{rank})$

return S_{scores}

Algorithm 5: Graph Attention Network (GAT) for Sentence Ranking

3.5 REDUNDANCY ELIMINATION

Redundancy elimination ensures that selected sentences provide diverse information without repetition. Given candidate sentences C, a reduced set R can be built by using various methodologies.

The algorithm 6 systematically identifies sentences and removes ones that are too similar to previously selected and ranked sentences. The threshold δ for similarity controls how aggressive the algorithm is in removing overlapping content. A higher threshold δ will remove more sentences, promoting greater redundancy.

Input: Set of candidate sentences C with embeddings

Output: Reduced set R

Initialize $R = \emptyset$.

for each sentence $c \in C$ **do**

  if $sim(c, r) < \delta$ for all $r \in R$ **then**

    Add c to R.

  end

end

Algorithm 6: Redundancy Elimination Algorithm

3.5.1 Cosine Similarity Filtering

The cosine similarity method is a simple, greedy, pair-wise approach to redundancy filtering. It iterates through the candidate sentences and comparing each new sentence to the set of sentences already selected. The cosine similarity

between the new candidate sentence's embedding vector and every selected sentence's vector is calculated. If the maximum similarity found exceeds the defined threshold (τ), the candidate sentence is deemed redundant and is removed.

3.5.2 Cluster-Based Selection

The cluster-based method is an instance-selection approach [10], [34] which can be used to remove duplicates. The method groups sentences into clusters based on their high similarity. For each cluster, only the most representative sentence is selected. This is effective at eliminating thematic redundancy by ensuring only one sentence is chosen to represent each distinct topic present.

3.5.3 Maximal Marginal Relevance

MMR simultaneously optimizes for two opposing criteria: relevance to the original document and diversity among the selected sentences. MMR is highly effective because it directly addresses the trade-off inherent in summarization: maintaining high content coverage while ensuring the summary is highly diverse and non-repetitive. Here, λ ($0 \leq \lambda \leq 1$) controls the balance between relevance and redundancy, higher λ favors relevance, while lower values favor diversity.

$$\mathrm{MMR}(S_i) = \lambda \cdot \mathrm{Relevance}(S_i) - (1-\lambda) \cdot \mathrm{Redundancy}(S_i, \mathrm{R}) \tag{3.1}$$

3.6 INFORMATION SELECTION

After redundancy elimination, sentences R are ranked by structural and semantic importance [15]. For each node v_i in G, a graph-based score $score_g(v_i)$ is computed using PageRank. An embedding-based score $score_e(v_i)$ is computed via cosine similarity. The final score is a weighted combination:

$$score(v_i) = \alpha \cdot score_g(v_i) + (1-\alpha) \cdot score_e(v_i). \tag{3.2}$$

Here, α ($0 \leq \alpha \leq 1$) determines the relative contribution of structural (graph-based) and semantic (embedding-based) importance in ranking sentences. Top-k ranked sentences are selected as the final summary S_{final}.

Input: Reduced set R, Graph G, Sentence embeddings E

Output: Final summary sentences S_{final}

for each sentence $s \in R$ **do**

  Compute graph-based importance score $score_g(v_i)$ for each node using PageRank.

  Compute embedding-based importance score $score_e(v_i)$ for each node from semantic similarity.

  Combine scores: $score(v_i) = \alpha \cdot score_g(v_i) + (1-\alpha) \cdot score_e(v_i)$.

end

Rank R by $score(v_i)$.

Select top-k ranked sentences $\rightarrow S_{final}$.

Return S_{final}.

Algorithm 7: Information Selection Algorithm

The Information Selection Algorithm 7 ranks sentences based on a combination of importance based on graph structure and semantic importance based on embeddings. The weighting factor α governs the relative influence of structural importance (PageRank score) versus semantic importance (embedding similarity). A balanced α ensures that both contextual significance and sentence meaning contribute to the final ranking.

Fig. 3.5 illustrates the step-by-step process of redundancy elimination and information selection in summarization. From an initial set of 10 input sentences, redundant or repetitive ones are removed to retain 7 key statements, and finally, the most informative 4 sentences are selected to form the concise summary.

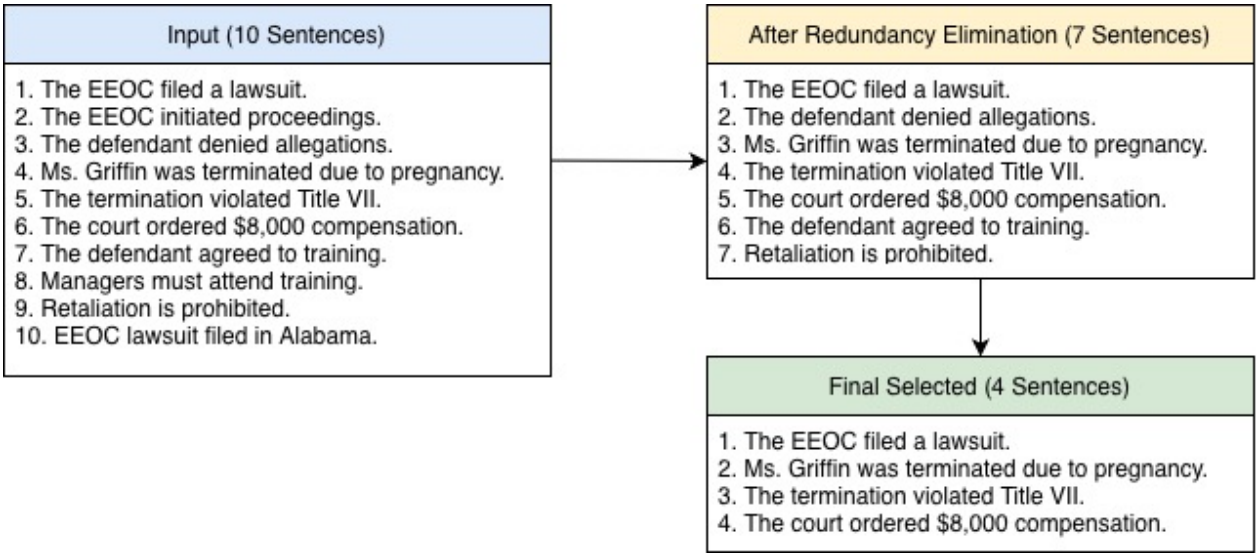

FIGURE 3.5: Example of redundancy elimination and information selection

3.7 TEMPORAL ORDERING

Chronological ordering arranges events in their temporal sequence using explicit dates or linguistic cues. In legal summarization, it preserves the procedural

narrative of filings and rulings. Methods such as [3], [14] capture implicit event orderings in unstructured texts. This is implemented using *dateparser* library. Fig. 3.6 illustrates the chronological ordering of legal events by sorting timestamped records, transforming an unordered input sequence into a temporally consistent timeline.

Input:

```
[
  {"timestamp": "2021-03-12T10:15:00", "event": "Defendant filed an
appeal"},
  {"timestamp": "2021-01-05T09:00:00", "event": "Court issued initial
ruling"},
  {"timestamp": "2021-02-28T14:30:00", "event": "Plaintiff submitted
additional evidence"}
]
```

Output:

```
[
  {"timestamp": "2021-01-05T09:00:00", "event": "Court issued initial
ruling"},
  {"timestamp": "2021-02-28T14:30:00", "event": "Plaintiff submitted
additional evidence"},
  {"timestamp": "2021-03-12T10:15:00", "event": "Defendant filed an
appeal"}
]
```

FIGURE 3.6: Example for Temporal Ordering

$$S_{sel} = \text{Order}\big(\text{TopK}(R_s, score, k)\big) \tag{3.3}$$

R_s denotes the redundancy-filtered candidate sentence set, *score* represents the combined structural and semantic importance score assigned to each sentence, and k is the desired summary length. The function $\text{TopK}(\cdot)$ selects the top k most relevant sentences, and $\text{Order}(\cdot)$ arranges them chronologically based on temporal cues extracted from the legal text.

3.8 FACT VERIFICATION

Fact verification determines whether a claim is supported, refuted, or unverifiable against trusted evidence. Given a claim c and evidence set E, the classifier outputs $\{\textit{Supported}, \textit{Refuted}, \textit{Not Enough Info}\}$. There has been research into using catchphrase identification in predicting the relationship between case law, and this work appears to be helpful in retrieving previously decided legal cases. In addition, the use of domain-specific markers will allow for the retrieval of case law relevant to the legal domain through the use of contextual references [25].

$$S_{final} = \text{Verify}(S_{abs}, E), \quad \text{label}(c_i) \in \{\text{Supported, Refuted, NEI}\} \tag{3.4}$$

The table 3.1 contains examples of fact verification, where each claim is checked against relevant extracted evidence, to see if the claim is factually consistent with the evidence. The model classifies statements as Supported, Refuted, or Not Enough Info depending on whether the claim and the evidence match.

TABLE 3.1: Example of fact verification.

Claim	**Evidence**	**Label**
The Supreme Court overturned the lower court's ruling on 12 March 2021.	On 12 March 2021, the Supreme Court upheld the decision.	Refuted
The defendant was acquitted on 5 July 2020.	The judgment issued on 5 July 2020 states: "The defendant is acquitted of all charges."	Supported
The case was filed in February 2019.	No filing date mentioned in available evidence.	Not Enough Info

3.9 SENTENCE SELECTION USING GCN AND GAT

The GNN approaches implement a graph-based extractive summarization pipeline. The complexity of converting the raw text document into a structured graph, as well as the calculation of initial sentence features, is performed before this stage. The GCN and GAT models are then used to refine these features based on the learned document topology.

Both implementations start by loading a pre constructed document graph. The model loads a graph object, where nodes represent sentences, and edges represent semantic connections. This serves as the starting point for the GNN's learning process. It then refines the initial features by propagating information across the graph's adjacency matrix.

The refined output feature vector from the final GNN layer is passed through a simple linear head to predict a continuous salience score for every sentence. Sentences are scored and ranked based on their predicted salience. Multiple lengths are evaluated against the ground truth reference using a metric like ROUGE-L. The K value that yields the highest ROUGE-L score is chosen as the final K value. The top K sentences are selected to form the final summary. The difference between the two approaches lies entirely in how they perform feature aggregation.

- **Graph Convolutional Network (GCN)**: Performs normalized feature aggregation where sentences gather information from neighbors based on a static graph structure. All neighbors contribute equally to the central node's

representation, and sentences are ultimately ranked by predicted salience scores.

- **Graph Attention Network (GAT)**: Utilizes a self-attention mechanism to assign variable weights to neighboring nodes during propagation. This prioritizes highly relevant connections for a more nuanced structural representation. The output length is optimized by selecting a K value that maximizes the target quality metric.

3.10 SUMMARIZATION

Summarization leverages different model classes to address sequence length, factual accuracy, and domain specificity. Transformer models [13] are widely used. Encoder–decoder models (e.g., BART, T5) [18] excel in abstraction but face input length limits. Long-document transformers (e.g., Longformer, LED) address length constraints. Hybrid approaches combine extraction and abstraction [8]. Domain-specific models such as Legal-BERT enhance legal reasoning. Open-source LLMs like LLaMA and Mistral [4], [23] can be used to offer scalability and long-context reasoning. These models can be integrated with graph-based frameworks for accurate, coherent legal summaries. Significant improvements have been made in recent years in transformer-based methods for summarising long-form documents, such as T5 and very large language models (e.g., Gemini). These advances have resulted in increased abstraction and coherence at the level of the long-form summary [24].

- **Encoder-Decoder Transformers:** Decoder only models have been used such as in [18], but Encoder–decoder transformer models (e.g., BART, T5) are better suited for abstractive summarization. However, they are constrained by input length limits, though recent work highlights how LLMs can enhance legal document analysis and summarization pipelines.

- **Long Document Transformers:** To overcome input size constraints, long-document transformers (e.g., LED) extend the transformer architecture with sparse or hierarchical attention mechanisms, enabling the processing of lengthy legal texts while capturing dependencies across multiple documents.

- **Hybrid Techniques:** Hybrid techniques that combine extraction and abstraction have also been shown to improve summarization of judgment documents in practice [8]. In parallel, region-specific transformer ensembles and transfer learning approaches further adapt summarization systems to jurisdictional and corpus-specific requirements.

- **Domain-Specific Models:** Additionally, domain-specific models (e.g., Legal-Pegasus) incorporate legal knowledge, discourse structures, and argumentation patterns to produce summaries that align with legal reasoning and terminology.

The core value of GNNs is their ability to capture global dependencies and document flow, which are critical for summarization. They allow the embedding of each sentence to be informed by its entire neighborhood and the document structure, creating richer, contextualized representations, leading to enhanced summarization.

CHAPTER 4

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

4.1 DATASET DESCRIPTION

The dataset that is used here is Multi-LexSum [27] consisting of 9,280 legal case summaries, created to support research on summarization of civil rights litigation. It is distinct in providing summaries at multiple levels of granularity — from a one-sentence "extreme" summary, to paragraph-length summaries, up to long, multi-paragraph summaries (over 500 words) for each case. Each case in the dataset includes multiple source documents related to civil rights lawsuits: complaints, court opinions, orders, settlement agreements, etc. The summaries are expert-authored, following detailed guidelines to ensure high quality. This makes Multi-LexSum more reliable than many automatically or semi-automatically constructed summarization datasets. It is thus a good benchmark for long document summarization, multi-document summarization, and for controllable summarization.
Dataset link: https://multilexsum.github.io/

Figures 4.1 and 4.2 illustrate examples from the proposed MultiLex-dataset, showcasing representative legal documents (Fig. 4.1) and their corresponding abstractive summaries (Fig. 4.2). These samples highlight the structural and linguistic diversity present across cases in the dataset.

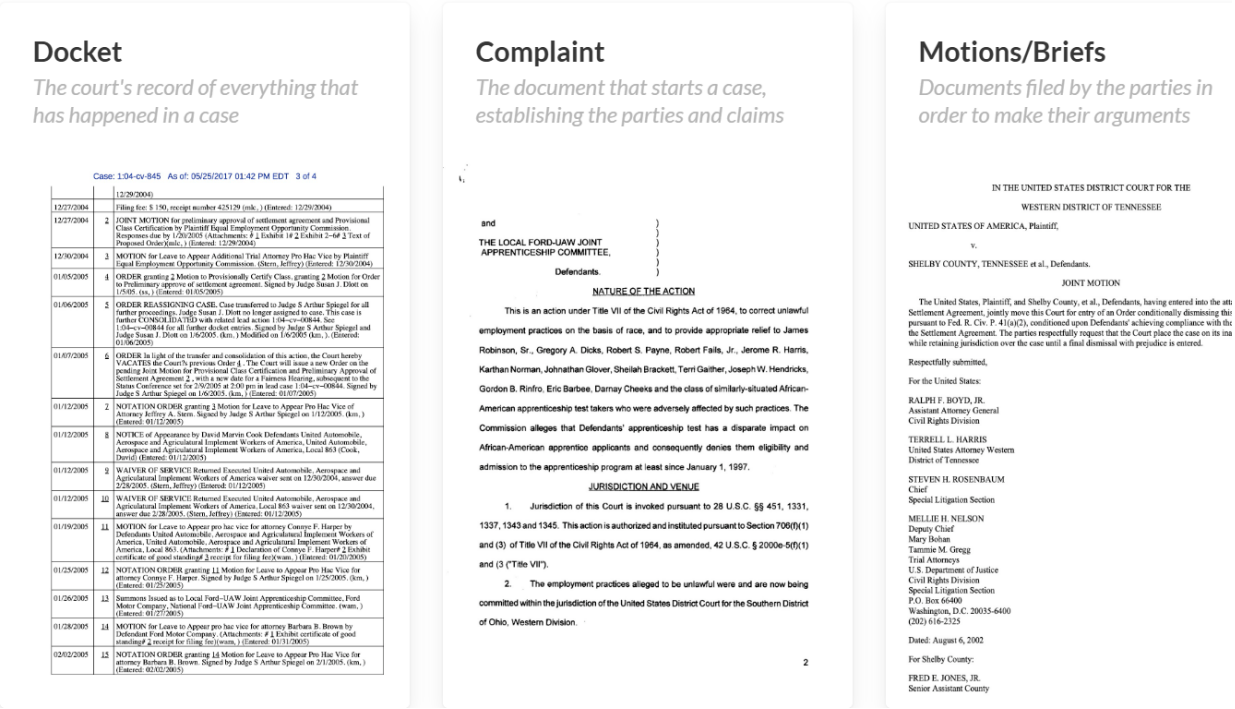

FIGURE 4.1: Sample documents from MultiLex-dataset

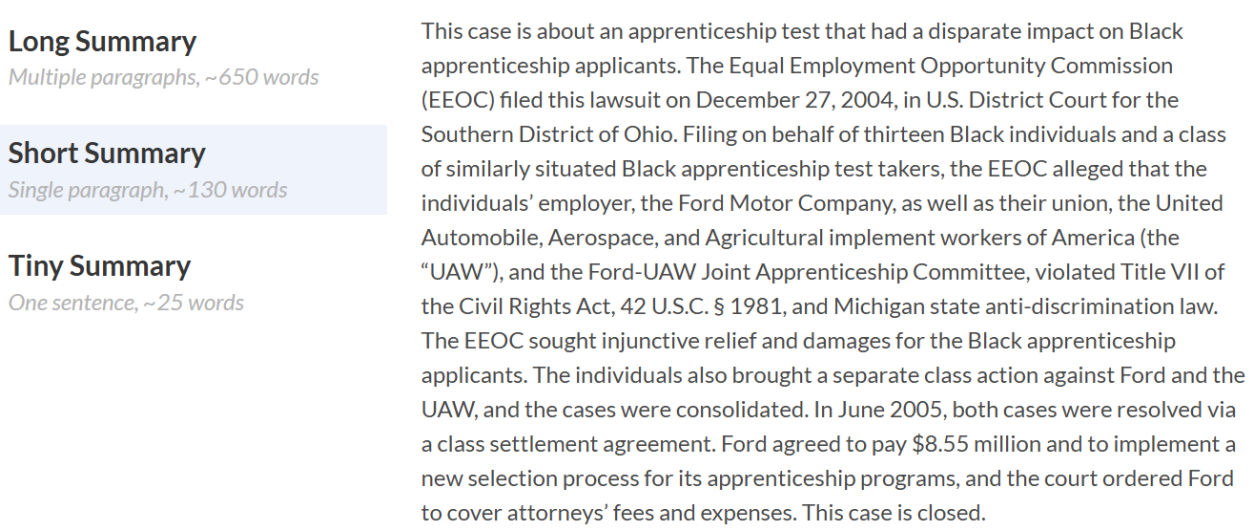

FIGURE 4.2: Sample summaries from MultiLex-dataset

To ensure that the developed model works for Indian legal documents as well, we are collecting data to create a dataset suitable for multi-document summarization. The documents are collected from https://www.sci.gov.in/case-status-case-no/ the Supreme Court of India website. The dataset consists of several cases, and each case consists of multiple documents. We are in the process of collaborating with professionals in the law domain to produce gold summaries for these cases.

4.2 ECOSYSTEM

This section details the technical environment in which the project was developed and is designed to operate. It covers the project's dependency structure, the physical infrastructure, and the core technological stack used across the pipeline stages. The pipeline is architecturally separated into distinct, loosely coupled stages: Preprocessing, Argument Mining, Graph Construction, Information Selection and Summary Generation. This modularity facilitates unit testing and independent scaling. Input documents are processed sequentially through the pipeline, with data passed between modules as serialized data structures like JSON and Pickle files.

4.2.1 Hardware Specification

The hardware requirements are split between the demanding Training phase and the less demanding Execution phase.

Due to the computational intensity of Natural Language Processing (NLP) models, dedicated hardware was necessary.

- **Processor (CPU):** Intel Core i7 (8th Gen or higher) or equivalent AMD for general pipeline execution.
- **GPU:** NVIDIA Tesla T4 or equivalent (minimum 15 GB VRAM). This is crucial for acceleration of Graph construction, pre-training LLMs with legal data, and Summarization.
- **RAM:** 12 GB RAM. This is used for handling large in-memory data structures like graph matrices and large document embeddings.
- **Storage:** 100 GB disk space for fast I/O for reading large datasets and models.

Once trained, the model requires minimal hardware for generating summaries, focusing on fast CPU performance.

4.2.2 Software Specification

This section lists the essential software, libraries, and frameworks utilized to build and run the multi-document summarization pipeline. The use of specific versions is crucial for reproducibility.

Key Dependencies for Reproducibility: All specific software library versions are precisely managed using a documented environment file (e.g., `requirements.txt`) to ensure the project can be reproduced with high fidelity.

TABLE 4.1: Software Specifications

Category	Component/Library	Usage in Pipeline Stage
Core Language	Python 3.9+	Main development language for all scripts and modules.
NLP Framework	PyTorch / TensorFlow	Argument Mining model implementation and execution.
NLP Libraries	Hugging Face Transformers	Accessing pre-trained Language Models for embeddings/annotation.
NLP Libraries	spaCy / NLTK	Preprocessing (Tokenization, POS tagging, Named Entity Recognition).
Graph Processing	NetworkX	Graph Construction and implementing graph-based ranking (e.g., PageRank).
Data Handling	NumPy, Pandas	Efficient numerical computation and tabular data management.

4.3 Evaluation Metrics

The framework's performance was assessed using lexical and semantic overlap metrics.

ROUGE-1 (Unigram Overlap):

$$ROUGE_1 = \frac{\sum_{gram_1 \in Y} \min(\text{Count}(gram_1 \in X), \text{Count}(gram_1 \in Y))}{\sum_{gram_1 \in Y} \text{Count}(gram_1 \in Y)}$$

Measures the overlap of single words between the generated (X) and reference (Y) summaries, capturing basic content match.

ROUGE-2 (Bigram Overlap):

$$ROUGE_2 = \frac{\sum_{gram_2 \in Y} \min(\text{Count}(gram_2 \in X), \text{Count}(gram_2 \in Y))}{\sum_{gram_2 \in Y} \text{Count}(gram_2 \in Y)}$$

Measures the overlap of contiguous word pairs (bigrams), which assesses local fluency and phrasing similarity.

ROUGE-L (Longest Common Subsequence):

$$ROUGE_L = \frac{LCS(X,Y)}{\max(|X|,|Y|)}$$

ROUGE-L assesses the similarity between a generated summary and a reference summary based on the longest matching contiguous sequence of words. It captures content similarity and fluency based on the structure of the sentences rather than identical words.

BERTScore-F1:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

BERTScore-F1 assesses semantic similarity between generated and reference summaries employing contextual embeddings from pretrained language models via precision, recall, and F1-score to understand how well the generated content preserves meaning beyond surface-level word matching.

4.4 CONDUCTION OF EXPERIMENTS

To evaluate the performance of our graph-based representation, we experimented with varying similarity thresholds ranging from 0.5 to 0.9. For each threshold, key network metrics were computed. The results show a sharp decline in density and centrality values between thresholds 0.5 and 0.6, followed by a gradual stabilization. This indicates that weaker semantic links are filtered out early, leaving only stronger and more reliable connections. The number of edges decreases significantly with higher thresholds, implying that thresholding primarily prunes noisy inter-sentence connections. Thus, a threshold value between 0.65–0.75 can be chosen as an optimal threshold for edge connections in the graph. It allows the filtering of less meaningful connections without excessively reducing the number of edges, which could make the graph too sparse for effective analysis. Fig. 4.3 illustrates the effect of varying graph threshold values on key structural metrics. A stable threshold value is selected where these metrics stabilize, ensuring an optimal balance between connectivity and sparsity.

Table 4.2 presents a quantitative analysis of graph properties across different threshold values. As the threshold increases, the number of edges, density, centrality, and clustering coefficients decrease, indicating a progressive sparsification of the graph structure.

Quantitative analysis verified that even though the three redundancy removal methods: Maximal Marginal Relevance (MMR), Clustering, and Cosine Similarity, resulted in the same compression rate (26 preserved nodes from 184 original nodes), they did not generate strictly identical summaries. Thus, deeper content comparison is warranted. To assess how different they actually were, we

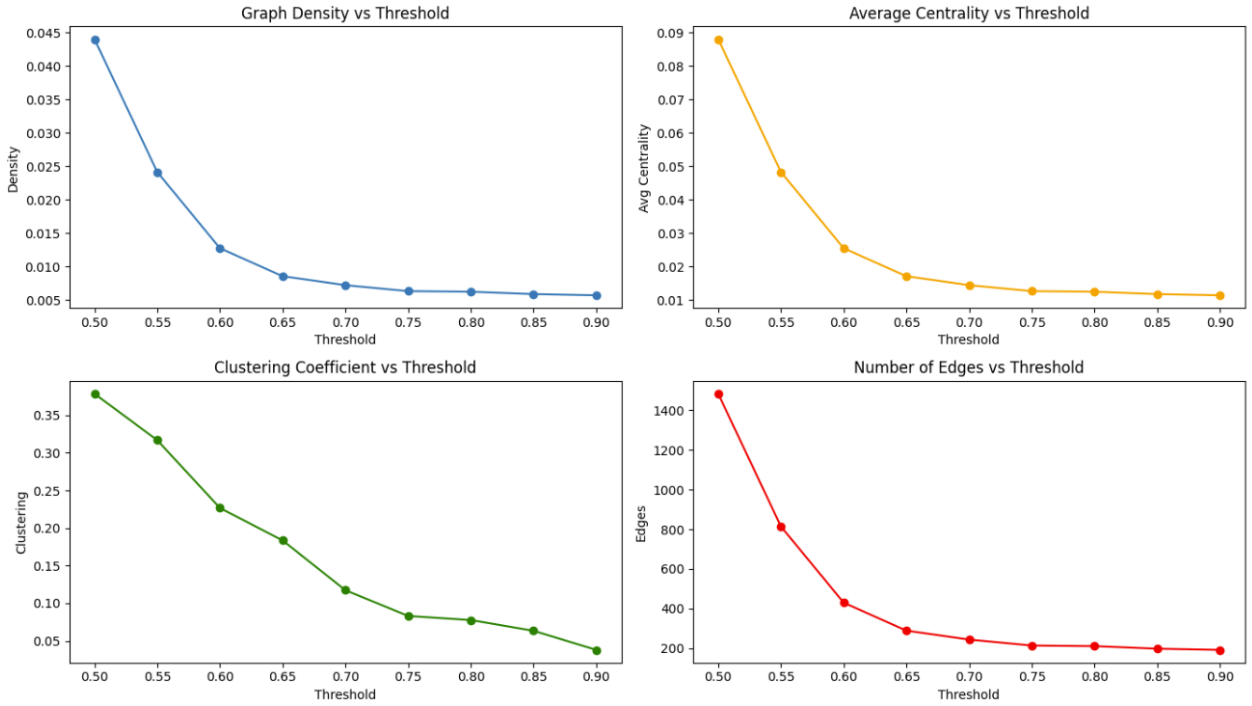

FIGURE 4.3: Analysis of varying Graph Threshold values

TABLE 4.2: Analysis of Graph Threshold values.

Threshold	Nodes	Edges	Density	Avg Centrality	Clustering
0.5	184	1483	0.044	0.088	0.379
0.55	184	814	0.024	0.048	0.317
0.6	184	430	0.013	0.026	0.227
0.65	184	289	0.009	0.017	0.183
0.7	184	244	0.007	0.014	0.117
0.75	184	214	0.006	0.013	0.083
0.8	184	211	0.006	0.013	0.078
0.85	184	198	0.006	0.012	0.063
0.9	184	192	0.006	0.011	0.038

applied the pairwise F1-score for set overlap, an established measure of content agreement based on the node IDs it retained.

The minimum F1 score for overlap (shown in Fig. 4.4) was 0.8846 indicating that even the ”most different” pair of summaries retained over 88.4% of the same sentences. This high degree of retention empirically demonstrates that each of the three methods was able to identify and subsequently retain the same substantive core units of argument and sequence of events. Thus, if less than 12% disagreement of content is valid disagreement in comparison arguments, then we consider it functionally insignificant for producing valid legal summary documents.

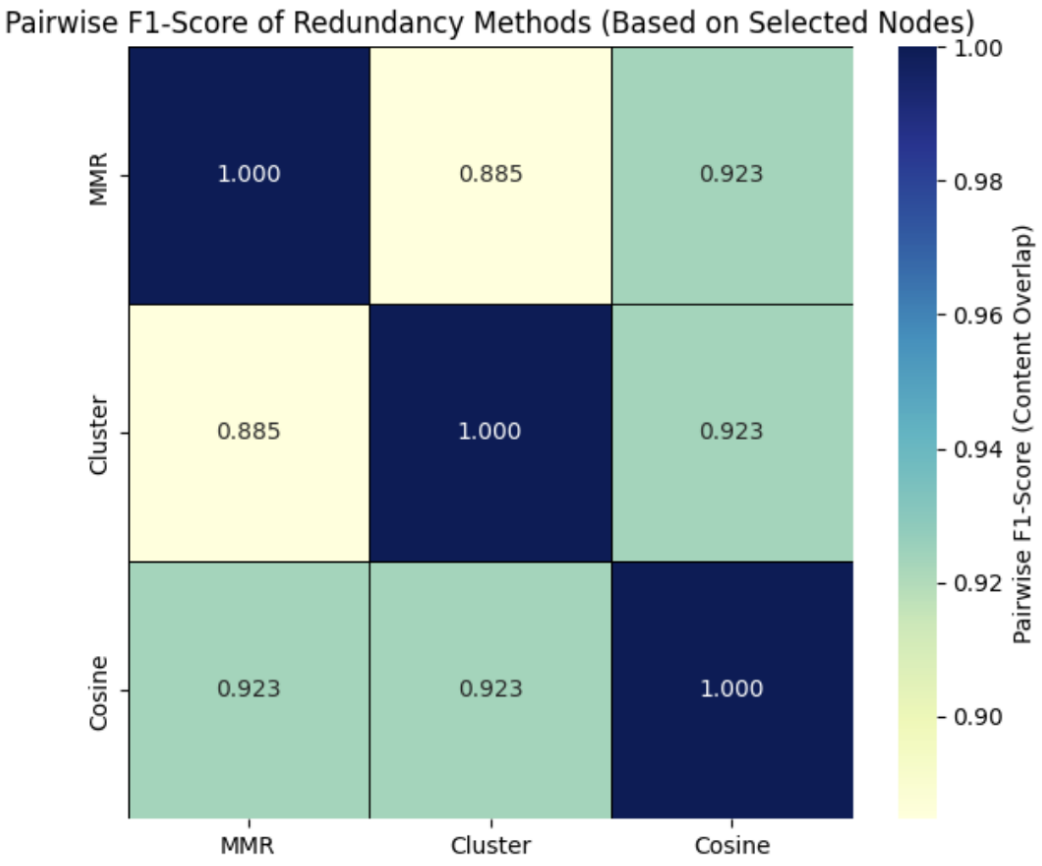

FIGURE 4.4: F1 Score of Redundancy Methods

As the redundancy threshold increases from 0.7 to 0.9, the percentage of content reduced decreases significantly, from over 22% to just over 6%. However, the ROUGE and BERTSCORE metrics [31], which measure the quality of the preserved content, consistently improve and then plateau. A threshold of 0.85 strikes the optimal balance, providing a reasonable 8.70% reduction in content, making it a well-rounded choice for ensuring conciseness and accuracy.

Table 4.3 shows the impact of varying redundancy elimination thresholds on summary quality and reduction rate. Higher thresholds result in fewer sentence reductions but improved ROUGE and BERTScore values, indicating better content preservation.

TABLE 4.3: Analysis of Redundancy Elimination Threshold values.

Threshold	**Original**	**Selected**	**Reduction (%)**	**ROUGE-1**	**ROUGE-L**	**BERTScore-F1**
0.7	184	142	22.83	0.877	0.877	0.951
0.75	184	156	15.22	0.927	0.927	1.000
0.8	184	164	10.87	0.947	0.947	1.000
0.85	184	168	8.70	0.959	0.959	1.000
0.9	184	172	6.52	0.965	0.965	1.000

The analysis for the short summary focused on smaller K values which is shown in Table 4.4, spanning $K = 3$ to $K = 10$. The optimal value was found to be $K = 5$, which resulted in the highest ROUGE-L score of 0.1633. For a short summary, a low K value is expected to be optimal, as the goal is conciseness. $K = 5$ demonstrates the maximum ability to select the five most critical, non-redundant sentences that align structurally with the reference. Increasing K beyond this point, results in a substantial drop in ROUGE-L, confirming that including too many sentences dilutes the core information.

TABLE 4.4: Evaluation of Candidate K Values for Short Summaries

K Value	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
3	0.3015	0.0881	0.1619	0.8292
5	**0.3150**	**0.0910**	**0.1633**	**0.8350**
7	0.2912	0.0820	0.1511	0.8287
10	0.2574	0.0735	0.1342	0.8250

The evaluation of candidate K values (number of sentences selected) for the long summary spanned from $K = 10$ to $K = 75$ is denoted in Table 4.5. The optimal value was determined to be $K = 60$, yielding the highest ROUGE-L score of 0.1626.

TABLE 4.5: Evaluation of Candidate K Values for Long Summaries

K Value	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
10	0.1250	0.0240	0.0714	0.8123
15	0.1250	0.0240	0.0714	0.8123
20	0.1162	0.0246	0.0612	0.8105
25	0.0743	0.0187	0.0619	0.8093
30	0.1520	0.0235	0.0877	0.8130
35	0.2165	0.0363	0.1186	0.8161
40	0.2078	0.0574	0.1299	0.8171
45	0.1872	0.0591	0.1283	0.8065
50	0.2767	0.0780	0.1359	0.8294
55	0.2750	0.0804	0.1350	0.8384
60	**0.3253**	**0.0927**	**0.1626**	**0.8192**
65	0.3053	0.0889	0.1549	0.8194
70	0.3163	0.0895	0.1381	0.8169
75	0.2986	0.0682	0.1493	0.8184

For long summary, optimal selection reflects the point where the benefit of including additional unique information (as shown by the high ROUGE-1 score of 0.3253) is maximized before redundancy begins to significantly degrade the overall coherence metrics. The general trend shows a substantial performance

increase from $K = 25$ (ROUGE-L=0.0619) up to $K = 60$, after which ROUGE-L begins to decline, indicating that $K = 60$ provides the best balance between informativeness and structural integrity for the longer summary format.

4.5 ABLATION STUDIES

4.5.1 Without Important Information Selection Step

When selection is removed, the input becomes a longer set of sentences that passed the redundancy filter. It is observed to be approximately double the size compared to the input after selecting important information. The large, consistent decrease in BERTScore-F1 (ranging from -0.0613 to -0.0817) as shown in Table 4.6 indicates that when the model is forced to summarize the entire document excepting redundant content, the resulting summaries lose semantic coherence and relevance to the ground truth because the entire content is summarized rather than just the important parts.

TABLE 4.6: Mean Metric Difference Without Important Selection vs Original

Threshold	Long Summaries				Short Summaries			
	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
0.65	**+0.0270**	+0.0086	+0.0174	−0.0613	+0.0046	+0.0128	+0.0195	−0.0679
0.70	+0.0160	−0.0027	+0.0028	−0.0744	−0.0308	+0.0017	−0.0064	−0.0817
0.75	+0.0011	−0.0053	−0.0027	−0.0669	−0.0113	−0.0086	−0.0017	−0.0712

While ROUGE scores were occasionally positive, this gain came at the severe cost of semantic accuracy. This affirms that the selection step is necessary to reduce noise and focus the model's attention on the most critical information, leading to semantically superior output despite having fewer input sentences.

4.5.2 Without Redundancy Elimination Step

When redundancy elimination is removed, the final input set is much larger because it includes sentences that convey duplicate information or are semantically too close to each other. It is observed to be twice as large compared to the input set after eliminating redundancy. The results in Table 4.7 show a classic trade-off: The system achieved higher ROUGE scores by including redundant content. However, this redundant content led to a substantial drop in BERTScore-F1 (negative differences of -0.0602 to -0.0790). This means the resulting summaries were less focused and semantically inferior to the gold standard.

TABLE 4.7: Mean Metric Difference Without Redundancy Elimination vs Original

Threshold	**Long Summaries**				**Short Summaries**			
	ROUGE-1	**ROUGE-2**	**ROUGE-L**	**BERTScore-F1**	**ROUGE-1**	**ROUGE-2**	**ROUGE-L**	**BERTScore-F1**
0.65	**+0.0722**	+0.0082	+0.0223	−0.0639	−0.0200	−0.0029	−0.0045	−0.0790
0.70	+0.0514	+0.0190	+0.0252	−0.0632	+0.0044	+0.0090	+0.0078	−0.0770
0.75	+0.0302	+0.0020	+0.0076	−0.0602	−0.0008	+0.0014	+0.0079	−0.0685

The elimination step successfully discards noise, ensuring the final summary focuses on non-redundant, high-value information, thus justifying the slight ROUGE sacrifice for a large gain in semantic quality. Hence, the redundancy step is essential for enforcing diversity and conciseness. Removing it increases the workload for the summarizer with poor-quality, repetitive data, which degrades the semantic quality of the final output.

4.5.3 Without Graph Neural Networks

The results clearly demonstrate that adding a GNN layer consistently improves the performance of summarization across lengths. The integration of GNNs into the

summarization pipeline provides a measurable performance uplift by transforming the way models process dependencies within legal texts. It allows the model to capture structural hierarchies that standard sequential transformers often overlook.

As shown in Table 4.8, this is evident across summarization models. While some models showed minor regressions, the overall trend suggests that graph-based embeddings provide a richer, more relational feature set that helps the transformer navigate the problem of scattered information in multi-document legal summarization.

TABLE 4.8: Impact of GNN on Simple Argument Graph

Model	**Long Summaries**			**Short Summaries**		
	Base	**GCN**	**(Δ)**	**Base**	**GCN**	**(Δ)**
FLAN-T5	**0.1184**	**0.1920**	**+0.0736**	**0.1552**	**0.2109**	**+0.0557**
T5-Small-Billsum	0.1414	0.1418	+0.0004	0.1486	0.1843	+0.0357
Long-T5	0.1219	0.1146	-0.0073	0.1514	0.1820	+0.0306
T5-Base	0.1519	0.1821	+0.0302	0.1594	0.1561	-0.0033
LED-Base	0.1558	0.1744	+0.0186	0.1146	0.1663	+0.0517
BART-Large-CNN	0.1710	0.1845	+0.0135	0.1432	0.1854	+0.0422
Legal-LED	0.1844	0.1929	+0.0085	0.1617	0.1699	+0.0082
Legal-PEGASUS	0.1645	0.1864	+0.0219	0.1420	0.1869	+0.0449
PEGASUS-XSum	0.1640	0.1772	+0.0132	0.2190	0.2234	+0.0044
PEGASUS-Billsum	0.1958	0.1856	-0.0102	0.1656	0.1922	+0.0266

In comparison to non-graph based approaches, models using graph based neural network layers show statistically significant results in terms of both accuracy and performance over various types of model, as illustrated in Figure 4.5. This is particularly true for shorter summaries, where graph based representations enhance both the semantic coherency and contextual understanding of the summaries.

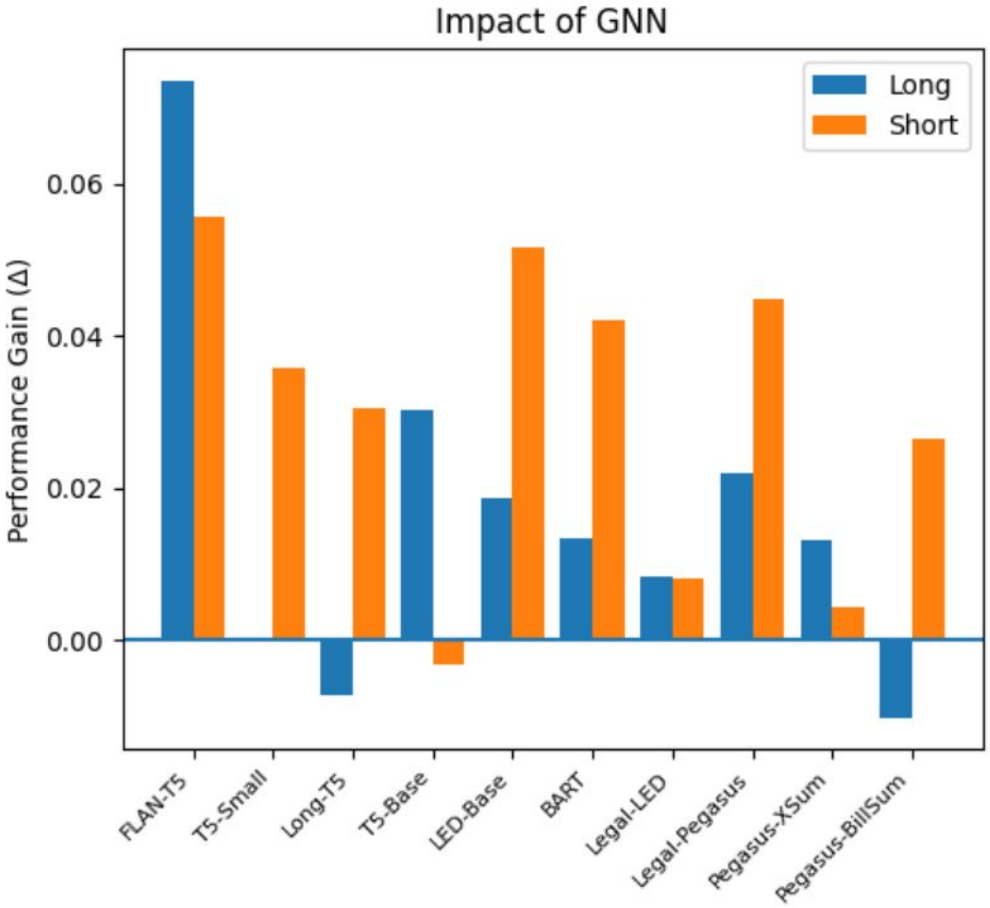

FIGURE 4.5: Impact of GNN on summatization models

4.5.4 Error Analysis

Models using general architectures (BART, T5) created using data from general sources have major limitations when trying to summarize legal documents. Since the structure of legal discourse differs from that of other types of writing, these models struggle. The result is a loss in the nuances present in the law, and also an incomplete summary. For instance, in a civil rights case summary, a sentence such as *"The court held that the defendant violated the petitioner's fundamental rights under Article 21"* may be simplified by a general model to *"The court ruled against the defendant"*, thereby losing the constitutional context and legal

reasoning.

Small models such as T5 and its variants have limitations due to their size; therefore, they do not contain enough capacity to learn long-range dependencies and multi-entity relationships, resulting in an incomplete summary. For example, these algorithms frequently overlook significant intermediate events (such as earlier appeals or prior orders) when summarizing multi-document cases with many parties and timelines, leading to inadequate summaries that omit crucial procedural steps.

Domain-Specific models (Legal-Pegasus, Legal-LED) are shown to perform better in dealing with terms of law, citations, and argument structure; however, results across all thresholds and length of summaries are not consistently better than general-domain models. In some cases, general-domain models will have comparable or superior performance to legal-domain models, showing that pre-training alone does not provide enough performance without effective graph creation and sentence selection methods.For example, a domain-specific model correctly preserves a citation such as *"as established in Maneka Gandhi v. Union of India"*, whereas a general model may omit the citation entirely or paraphrase it incorrectly.

The use of thresholds such as $\tau = 0.75$ result in sparser graphs because it only connects highly similar nodes, leading to summaries that are not as information enriched. Compared to the other thresholds used, this resulted in a clear loss in summary quality, which was shown in the performance metrics. One such example, two sentences describing the same legal issue with different phrasing—*"violation of procedural fairness"* and *"denial of natural*

justice"—may not be linked at higher thresholds, causing one of them to be excluded and reducing informational completeness in the final summary.

Overall, errors stem from ambiguity in legal discourse and argument labeling, and model limitations in domain adaptation and long-context reasoning. The general conclusions about the errors made suggest that most of the limitations in performance are due to an interaction between the degree of connectivity of the graph structure, and the model's ability to generalize. In the future, we should work on adaptively setting graph thresholds, contextually aware removal of redundancy, and creating a hybrid between GNN and attention mechanisms in order to achieve even better temporal coherence and completeness of arguments.

4.6 RESULTS

4.6.1 Results using Phase-1 Pipeline

The Phase 1 starts with preprocessing the input data. After the input data has been cleaned and segmented into various components (sentences, clauses, etc.), argument mining is performed in order to identify and classify each component of the argument into their required classifications (claims, premises, evidence, and counter-arguments). The result of this will be used to create the Knowledge Graph, perform redundancy elimination, selecting Important Information, and create coherent summaries through an Abstractive summarization model based on the original data and the enriched graph. Phase 1 pipleline does not include the Graph Neural Networks (GNN) and Fact Verification module.

Graph-based modeling is employed to explicitly capture inter-sentence relationships in the Phase-1 pipeline. By representing sentences as nodes in a graph, the proposed approach enables structured information propagation for improved sentence ranking. It evaluates the effect of varying cosine similarity thresholds on redundancy reduction across different summarization models for both long and short summaries.

The consistent improvement of ROUGE and BERTScore metrics across models demonstrates the framework's capability to maintain both surface-level lexical overlap and deep semantic fidelity, preserving key legal arguments and contextual relationships despite a severe reduction of content.

The configuration thresholds of 0.65–0.75 consistently return the most balanced performance for both long and short summaries, offering cohesive summaries with strong factual accuracy and reduced redundancy. Furthermore, the analysis suggests that the inclusion of graph-theoretic attributes leads to more interpretable summary results.

The results further signify our hybrid approach of argument-based graph construction with semantic similarity filtering effectively filtered the irrelevant or repetitive content while enabling argumentative flow and case relevance. Table 4.9 reports corresponding ROUGE and BERTScore metrics across models and thresholds.

TABLE 4.9: Comparative Analysis of Cosine Similarity Redundancy Thresholds Across All Models for Long and Short Summaries (Final Phase-1 Results)

Model	Threshold	Long Summaries				Short Summaries			
		ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
allenai/led-base-16384	0.65	0.4050	0.0980	0.1453	0.8188	0.2362	0.0535	0.1067	0.8173
	0.70	0.3774	0.0947	0.1623	0.8285	0.2950	0.0653	0.1357	0.8338
	0.75	0.4063	0.0920	0.1599	0.8215	0.1993	0.0596	0.1014	0.8132
facebook/bart-large-cnn	0.65	0.3586	0.0800	0.1394	0.8171	0.2958	0.0647	0.1222	0.8163
	0.70	0.3770	0.1036	0.1786	0.8189	0.2812	0.0579	0.1406	0.8163
	0.75	0.3248	0.0839	0.1949	0.8172	0.2750	0.0420	0.1667	0.8199
google/flan-t5-base	0.65	0.2536	0.0433	0.1100	0.8034	0.2203	0.0356	0.1145	0.8034
	0.70	0.2096	0.0741	0.1246	0.8373	0.3580	0.1250	0.1728	0.8570
	0.75	0.1954	0.0578	0.1207	0.8129	0.3312	0.1032	0.1783	0.8276
google/long-t5-tglobal-base	0.65	0.1459	0.0306	0.1094	0.7989	0.2754	0.0441	0.1739	0.8048
	0.70	0.2663	0.0455	0.1256	0.8009	0.2705	0.0293	0.1449	0.8034
	0.75	0.2663	0.0505	0.1307	0.8011	0.2705	0.0293	0.1353	0.8031
google/pegasus-billsum	0.65	0.4203	0.0844	0.1821	0.8233	0.2842	0.0635	0.1632	0.8279
	0.70	**0.4474**	**0.1206**	**0.2137**	**0.8237**	0.2255	0.0443	0.1225	0.8198
	0.75	0.3271	0.0939	0.1916	0.8258	0.3207	0.0766	0.2110	0.8255
google/pegasus-xsum	0.65	0.3294	0.0946	0.1741	0.8329	0.3761	0.1121	0.2137	0.8554
	0.70	0.2393	0.0917	0.1595	0.8242	**0.3875**	**0.1519**	**0.2375**	**0.8454**
	0.75	0.2514	0.1099	0.1585	0.8369	0.4114	0.1618	0.2057	0.8525
nsi319/legal-led-base-16384	0.65	0.4331	0.1118	0.1847	0.8379	0.2471	0.0644	0.1373	0.8287
	0.70	0.3711	0.0663	0.1979	0.8306	0.2925	0.0411	0.1701	0.8319
	0.75	0.3739	0.0765	0.1706	0.8287	0.2778	0.0559	0.1778	0.8265
nsi319/legal-pegasus	0.65	0.3632	0.0791	0.1751	0.8261	0.3233	0.0606	0.1429	0.8441
	0.70	0.3537	0.1093	0.1633	0.8398	0.2880	0.0645	0.1440	0.8403
	0.75	0.3664	0.0866	0.1552	0.8371	0.3370	0.0517	0.1392	0.8386
stevhliu/t5-small-finetuned-billsum-ca_test	0.65	0.2600	0.0523	0.1324	0.8086	0.2759	0.0348	0.1552	0.8119
	0.70	0.2740	0.0676	0.1442	0.8171	0.2756	0.0628	0.1422	0.8281
	0.75	0.2905	0.0861	0.1476	0.8188	0.3057	0.0793	0.1485	0.8309
t5-base	0.65	0.3677	0.1055	0.1535	0.8196	0.2895	0.0596	0.1316	0.8221
	0.70	0.3271	0.0798	0.1542	0.8252	0.3629	0.1021	0.1688	0.8444
	0.75	0.3012	0.0844	0.1481	0.8297	0.3645	0.1038	0.1776	0.8413

4.6.2 Results using GNNs

To assess the impact of graph neural networks, we apply a Graph Convolutional Network (GCN) on top of the Phase-1 graph for sentence ranking. The performance obtained using GCN-based aggregation across multiple thresholds for both long and short summaries is reported in Table 4.10.

From the results, most models show clear ROUGE-L improvements over the Phase-1 pipeline for both long and short summaries, particularly at thresholds

0.65 and 0.70. This indicates that GCN-based global sentence aggregation effectively reduces redundancy while preserving salient content.

TABLE 4.10: GCN Results

Model	Threshold	Long Summaries				Short Summaries			
		ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
allenai/led-base-16384	0.65	0.4396	0.1088	0.1818	0.8198	0.3024	0.0623	0.1512	0.8228
	0.70	**0.4514**	**0.1342**	**0.1864**	**0.8240**	**0.3116**	**0.0803**	**0.1739**	**0.8185**
	0.75	0.3864	0.1037	0.1551	0.8126	0.3116	0.0803	0.1739	0.8185
facebook/bart-large-cnn	0.65	0.4123	0.1349	0.1901	0.8187	0.3529	0.0714	0.1647	0.8223
	0.70	**0.4018**	**0.1156**	**0.1684**	**0.8237**	**0.3804**	**0.0989**	**0.1957**	**0.8200**
	0.75	0.3897	0.1144	0.1949	0.8165	0.3804	0.0989	0.1957	0.8200
google/flan-t5-base	0.65	0.3953	0.1429	0.2134	0.8212	0.3226	0.0543	0.1828	0.8161
	0.70	0.4113	0.1364	0.2226	0.8230	0.2959	0.0958	0.2249	0.8352
	0.75	0.2073	0.0732	0.1401	0.8065	0.2959	0.0958	0.2249	0.8352
google/long-t5-tglobal-base	0.65	0.1321	0.0242	0.0961	0.7825	0.3402	0.0625	0.1856	0.8204
	0.70	0.2481	0.0571	0.1344	0.8010	0.3694	0.0727	0.1802	0.8249
	0.75	0.1970	0.0297	0.1133	0.8027	0.3694	0.0727	0.1802	0.8249
google/pegasus-billsum	0.65	0.3047	0.0741	0.1523	0.8221	0.4000	0.1064	0.2421	0.8330
	0.70	0.4420	0.1341	0.1910	0.8253	0.3636	0.0806	0.1673	0.8201
	0.75	0.3878	0.1444	0.2135	0.8316	0.3636	0.0806	0.1673	0.8201
google/pegasus-xsum	0.65	0.4148	0.1316	0.1921	0.8298	0.4192	0.1455	0.2635	0.8242
	0.70	0.3341	0.1295	0.1909	0.8290	0.3842	0.1029	0.2034	0.8201
	0.75	0.2343	0.0690	0.1486	0.8222	0.3842	0.1029	0.2034	0.8201
nsi319/legal-led-base-16384	0.65	0.4014	0.0998	0.1852	0.8247	0.3121	0.0857	0.1915	0.8366
	0.70	0.3756	0.1121	0.1997	0.8274	0.3045	0.0767	0.1592	0.8390
	0.75	0.4127	0.1115	0.1937	0.8283	0.3045	0.0767	0.1592	0.8390
nsi319/legal-pegasus	0.65	0.3720	0.0843	0.1640	0.8351	0.3522	0.1003	0.1728	0.8510
	0.70	0.4041	0.0911	0.2021	0.8360	0.3657	0.1053	0.1940	0.8565
	0.75	0.4372	0.1280	0.1931	0.8331	0.3657	0.1053	0.1940	0.8565
stevhliu/t5-small-finetuned-billsum-ca_test	0.65	0.2247	0.0847	0.1067	0.8061	0.3372	0.0588	0.1744	0.8221
	0.70	0.2159	0.1257	0.1648	0.8159	0.3314	0.0479	0.1893	0.8227
	0.75	0.2599	0.1013	0.1538	0.8073	0.3314	0.0479	0.1893	0.8227
t5-base	0.65	0.4171	0.1324	0.1925	0.8255	0.2454	0.0621	0.1472	0.8335
	0.70	**0.4131**	**0.1347**	**0.1859**	**0.8204**	**0.3210**	**0.0750**	**0.1605**	**0.8362**
	0.75	0.3129	0.0638	0.1678	0.8051	0.3210	0.0750	0.1605	0.8362

Further, we evaluate a Graph Attention Network (GAT) to capture fine-grained sentence importance through learned attention weights. The results obtained using GAT for long and short summaries are summarized in Table 4.11. The GAT results demonstrate further ROUGE-L improvements compared to GCN for certain models, especially in long summaries. The attention mechanism enables adaptive weighting of sentence relationships, leading to better identification of salient sentences. However, similar to GCN, higher thresholds introduce sensitivity and occasional performance drops for certain models.

TABLE 4.11: GAT Results

Model	Threshold	Long Summaries				Short Summaries			
		ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
allenai/led-base-16384	0.65	0.4182	0.1237	0.1984	0.8281	0.3007	0.0845	0.1748	0.8239
	0.70	0.3995	0.1015	0.1860	0.8180	0.2960	0.0945	0.1733	0.8197
	0.75	0.4306	0.1245	0.1770	0.8234	0.3116	0.0803	0.1739	0.8185
facebook/bart-large-cnn	0.65	0.3787	0.0769	0.1915	0.8206	0.3118	0.0652	0.1828	0.8290
	0.70	0.3839	0.1390	0.2143	0.8243	0.3448	0.0930	0.1954	0.8219
	0.75	0.3747	0.1313	0.2309	0.8321	0.3804	0.0989	0.1957	0.8200
google/flan-t5-base	0.65	0.3783	0.1131	0.1891	0.8254	0.2959	0.0958	0.2249	0.8352
	0.70	0.2937	0.0562	0.1445	0.8105	0.2976	0.0964	0.2262	0.8341
	0.75	0.3200	0.0567	0.1694	0.8106	0.2959	0.0958	0.2249	0.8352
google/long-t5-tglobal-base	0.65	0.0549	0.0050	0.0449	0.7412	0.2733	0.0629	0.1615	0.8329
	0.70	0.2788	0.0580	0.1298	0.8018	0.3095	0.1205	0.2262	0.8155
	0.75	0.3453	0.0846	0.1768	0.8093	0.3694	0.0727	0.1802	0.8249
google/pegasus-billsum	0.65	0.2837	0.0723	0.1322	0.7863	0.2871	0.0290	0.1340	0.8100
	0.70	**0.4596**	**0.1328**	**0.1985**	**0.8249**	0.3216	0.0406	0.1307	0.8242
	0.75	0.3953	0.1022	0.1526	0.8245	0.3636	0.0806	0.1673	0.8201
google/pegasus-xsum	0.65	0.4017	0.1258	0.1771	0.8319	0.3587	0.0879	0.1848	0.8189
	0.70	0.3826	0.0873	0.1696	0.8229	**0.3608**	**0.0625**	**0.1959**	**0.8420**
	0.75	0.3826	0.0873	0.1696	0.8229	0.3842	0.1029	0.2034	0.8201
nsi319/legal-led-base-16384	0.65	0.4061	0.1115	0.2031	0.8354	0.2797	0.0634	0.1608	0.8290
	0.70	0.4160	0.1245	0.2280	0.8471	0.2708	0.0559	0.1736	0.8316
	0.75	0.3915	0.1179	0.2064	0.8329	0.3045	0.0767	0.1592	0.8390
nsi319/legal-pegasus	0.65	0.3709	0.0998	0.1836	0.8347	0.2894	0.0583	0.1543	0.8385
	0.70	0.3992	0.1023	0.1996	0.8493	0.3101	0.0764	0.1835	0.8450
	0.75	0.4456	0.1195	0.1973	0.8473	0.3657	0.1053	0.1940	0.8565
stevhliu/t5-small-finetuned-billsum-ca_test	0.65	0.4315	0.0916	0.1669	0.8096	0.3282	0.1036	0.1949	0.8220
	0.70	0.3467	0.0714	0.1867	0.8104	0.3314	0.0479	0.1893	0.8227
	0.75	0.3763	0.1068	0.2045	0.8204	0.3314	0.0479	0.1893	0.8227
t5-base	0.65	0.2301	0.0661	0.1479	0.7718	0.2667	0.0736	0.1818	0.8280
	0.70	0.2314	0.0665	0.1598	0.7695	0.2909	0.0736	0.1697	0.8393
	0.75	0.2071	0.0767	0.1526	0.7696	0.3210	0.0750	0.1605	0.8362

Comparing the two graph neural networks, we can say that GCN is a better choice due its superior performance across granularity levels. It provides a stable and efficient outcome irrespective of model or threshold, forming a solid architecture for legal summarization tasks.

4.7 ANALYSIS OF RESULTS

The results demonstrate that argument mining, graph-based reasoning, and transformer-based summarization can be successfully integrated to produce fluent

and legally faithful summaries. Furthermore, these findings confirm that a hybrid framework which combines structural, semantic, and argumentative cues can yield summaries that are more relevant within the appropriate context as well as factually accurate than both purely extractive or abstractive methods.

The graph-based representation provides a natural mechanism for capturing inter-sentence and inter-document relationships that are critical for reconstructing legal narratives. To create edges in the graph that maintain semantically meaningful relationships and reduce noise in the graph, we enable optimal thresholds of 0.65-0.75. This decision promotes the interpretive power of the summarization. The stability of graph density and centrality metrics after achieving the threshold indicates that the model is able to distill out strong semantic dependencies, which supports the hypothesis that argument-aware connectivity improves flow of information and contextual embeddings propagated by the Graph Attention Network.

The experiment of eliminating redundancy found that a moderate threshold around 0.85 provided an acceptable balance of conciseness versus maintaining meaning. If too much compression is applied, crucial arguments or facts may be accidentally removed that are key components of any legal argument, where each claim and piece of evidence piece together how reasoned understanding is derived. The near perfect BERTScore results indicate the process of filtering redundancy nonetheless maintained the semantic core of the documents while enhancing the overall readability and focus on the issues in question. This demonstrates that redundancy detection effectively identified paraphrased or repetitive pieces of text, across related documents, and addressed a common challenge with more traditional extractive summarization systems.

The ablation studies performed by removing the redundancy elimination and important information selection steps individually show a consistent decrease in performance along with an increase in computation due to increased input size. This demonstrates the necessity of both steps in the pipeline.

The argument mining step worked to sub-segment the text before summarizing. The system classified the sentences into argumentative roles: claims, evidence, rules, and background, which meant the summary was grounded in legal reasoning continuity. The summary then adhered to prior studies that argue for and use discourse segmentation and identification of rhetorical roles, as necessary and important components of summarizing law [22], [17]. However, contrary to prior studies, that use sentence embeddings or keyword heuristics in relation to those rhetorical roles, the combination of role-based labels, getting at the types of supportive elements in an argument, and that which encodes the discourse relations across sentences via graph attention, makes this frame more robust and fuller in two respects.

Temporal sequencing and fact-checking improved summary accuracy much further. Several conventional task models do not take into account a temporal progression which can cause summaries to be misplaced in the procedural sequence of the cases at hand. Our findings indicate that introducing modules for temporal reasoning helps maintain the lineage of events in legal cases, and elevated both coherency in the narrative and interpretable results. Therefore, these summaries are more factually precise and logically consistent.

Significantly, the strong performance of the model on the Multi-LexSum and Indian legal corpora highlights its potential for adaptability across jurisdictions. This cross-domain indication suggests the model approach is not restricted to any

linguistic or structural variability of legal writing, providing strong implications of use in multilingual and cross-legal systems. The explainability of graph-based reasoning as an artificial intelligence approach also indicates a good fit for legal decision-support tools that require interpretability.

To summarize, the discussion highlights that the proposed argument-aware, graph-based summarization model demonstrates a method for bridging gaps in existing summarization research dealing with legal texts, especially concerning argumentative coherence, temporal fidelity, and factual consistency. The model is a balanced approach to the integration of symbolic reasoning and deep contextual modeling; therefore, as it pertains to the future directions of legal natural language processing, it has shown promise.

4.8 IMPACT OF GRAPH NEURAL NETWORKS ON SUMMARIZATION PERFORMANCE

The comparative analysis between the Phase-1 pipeline and the Graph Neural Network (GNN) enhanced pipeline reveals a distinct and substantial improvement in summarization quality across the board. The most significant gains were observed in the generation of long summaries, where the introduction of graph structures enabled models to capture long-range dependencies that a simple linear thresholding mechanism could not. This suggests that the GNNs successfully model the complex structural context of legal documents. While the Phase-1 pipeline already provided competitive scores for short summaries, the application of GNNs further pushed the performance ceiling.

Regarding the specific neural architecture, the results indicate a general preference for the Graph Convolutional Network (GCN) across the majority of the evaluated models. It proved to be the superior structure for high-performing models, likely due to its efficient and uniform aggregation of neighborhood information which stabilizes the feature representations. However, the Graph Attention Network (GAT) demonstrated specific utility for certain models in the long summary setting. The graph models converge on similar redundancy optimizations at higher thresholds. Overall, the use of GCN with a knowledge graph threshold of 0.70 gave optimal results as visualized in Fig. 4.6.

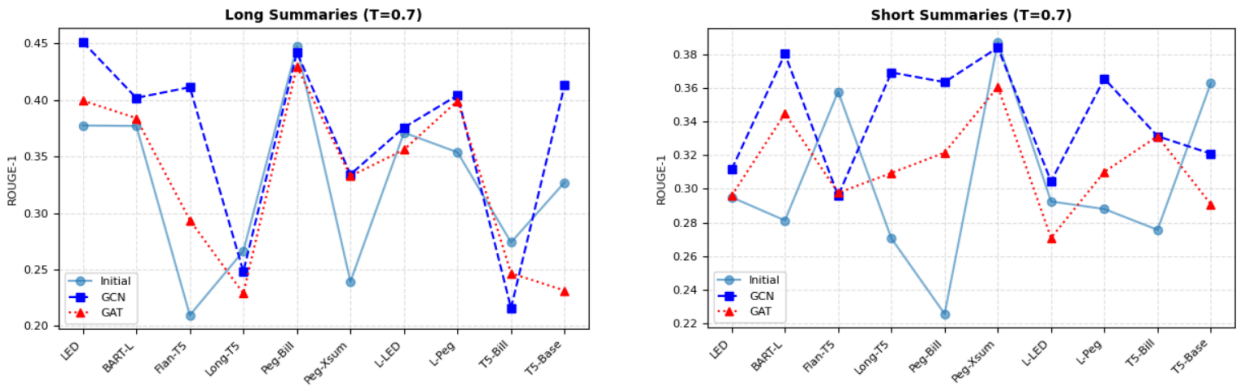

FIGURE 4.6: Summarization Performance across Pipelines

4.9 FACT VERIFICATION

Table 4.12 and Table 4.13, show that the different models display varying levels of faithfulness to the source content. We adopt a graded scoring scheme where neutral claims receive partial credit, as some statements are not explicitly supported but also not contradicted by the source, which is common in abstractive summarization.

The GCN summaries are stable, whereas the GAT summaries show more variance in terms of correctness. This suggests GCN is better at maintaining a baseline of factual accuracy.

TABLE 4.12: Fact Verification Results for GCN Architecture

Model	**Long Summary**			**Short Summary**		
	0.65	**0.70**	**0.75**	**0.65**	**0.70**	**0.75**
T5-Base	**0.632 (19)**	**0.719 (16)**	**0.786 (7)**	**0.50 (4)**	**0.75 (4)**	**0.75 (4)**
BART-Large-CNN	**0.833 (15)**	**0.618 (17)**	**0.889 (9)**	**0.75 (2)**	**0.833 (3)**	**0.833 (3)**
PEGASUS-XSum	0.60 (5)	0.667 (3)	0.333 (3)	0.50 (1)	0.50 (1)	0.50 (1)
LED-Base	**0.571 (14)**	**0.692 (13)**	**0.429 (14)**	**0.90 (5)**	**0.625 (4)**	**0.625 (4)**
Legal-PEGASUS	0.583 (6)	0.50 (6)	0.50 (5)	0.50 (7)	0.50 (5)	0.50 (5)
Long-T5	0.50 (1)	0.50 (1)	0.75 (2)	0.75 (2)	0.833 (3)	0.833 (3)
FLAN-T5	0.50 (6)	0.75 (8)	0.75 (2)	0.50 (2)	0.75 (2)	0.75 (2)
Legal-T5-Billsum	0.75 (2)	1.00 (3)	1.00 (3)	0.75 (2)	1.00 (2)	1.00 (2)
PEGASUS-Billsum	0.667 (3)	0.50 (5)	0.50 (2)	0.50 (1)	0.625 (4)	0.625 (4)
Legal-LED	0.438 (8)	0.45 (10)	0.50 (9)	0.40 (5)	0.50 (5)	0.50 (5)

Note: Values denote Fact Verification Score followed by (No. of Claims).

TABLE 4.13: Fact Verification Results for GAT Architecture

Model	**Long Summary**			**Short Summary**		
	0.65	**0.70**	**0.75**	**0.65**	**0.70**	**0.75**
T5-Base	**0.60 (5)**	**0.75 (4)**	**0.50 (5)**	**0.75 (4)**	**0.50 (4)**	**0.75 (4)**
BART-Large-CNN	**0.667 (6)**	**0.938 (8)**	**0.75 (2)**	**0.857 (7)**	**0.812 (8)**	**0.833 (3)**
PEGASUS-XSum	0.50 (1)	0.50 (1)	0.50 (1)	0.50 (1)	0.50 (1)	0.50 (1)
LED-Base	**0.538 (13)**	**0.536 (14)**	**0.625 (4)**	**0.70 (5)**	**0.50 (19)**	**0.625 (4)**
Legal-PEGASUS	0.50 (7)	0.50 (5)	0.55 (10)	0.50 (8)	0.50 (11)	0.50 (5)
Long-T5	0.50 (2)	0.50 (3)	0.75 (2)	1.00 (1)	0.50 (2)	0.833 (3)
FLAN-T5	0.50 (4)	1.00 (2)	0.75 (2)	0.75 (2)	0.333 (3)	0.75 (2)
Legal-T5-Billsum	0.50 (6)	0.25 (2)	0.75 (2)	0.75 (4)	0.333 (3)	1.00 (2)
PEGASUS-Billsum	0.70 (5)	0.625 (4)	0.333 (3)	0.50 (3)	0.333 (3)	0.625 (4)
Legal-LED	0.50 (6)	0.50 (5)	0.40 (5)	0.50 (6)	0.417 (6)	0.50 (5)

Note: Values indicate Fact Verification Score with (No. of Claims).

It can be observed faithfulness is inversely proportional to output length as most models showed a performance boost for short summaries. General-purpose models

currently outperform legally specialized models in this specific fact-verification task. This suggests that legal pre-training doesn't automatically translate to better logical faithfulness.

4.10 WORK ON INDIAN DATASET

The implementation of the summarization pipeline has been successfully completed and evaluated on the Indian legal dataset under development. The results on the Indian dataset as shown in Table 4.14 are up to par with the results obtained using the Multi-LexSum dataset, proving the robustness and legal generalization of the developed pipeline.

Achieving satisfactory performance metrics on this new corpus confirms that the foundational extractive approach is not limited to the original training data but is effective across diverse legal traditions and document structures. This cross-domain validation is a critical finding.

TABLE 4.14: Evaluation Results for Indian Dataset

Model	**Long Summaries**				**Short Summaries**			
	R-1	**R-2**	**R-L**	**BERT-F1**	**R-1**	**R-2**	**R-L**	**BERT-F1**
allenai/led-base-16384	**0.4488**	**0.1232**	**0.1920**	**0.8317**	**0.3651**	**0.1011**	**0.1958**	**0.8352**
facebook/bart-large-cnn	**0.4987**	**0.1146**	**0.2000**	**0.8336**	**0.3557**	**0.0717**	**0.2134**	**0.8398**
google/flan-t5-base	0.2952	0.0556	0.1328	0.8138	0.3465	0.0635	0.1890	0.8310
google/long-t5-tglobal-base	0.1109	0.0356	0.0576	0.8067	0.4132	0.0833	0.2149	0.8297
google/pegasus-billsum	0.2831	0.0740	0.1354	0.7863	0.3420	0.0915	0.1740	0.8240
google/pegasus-xsum	0.3972	0.1446	0.1722	0.8448	0.3945	0.1115	0.1869	0.8483
nsi319/legal-led-base-16384	0.4240	0.1119	0.1841	0.8252	0.3825	0.0934	0.2077	0.8399
nsi319/legal-pegasus	0.3754	0.0910	0.1825	0.8254	0.3641	0.1023	0.1912	0.8420
stevhliu/t5-small-billsum	0.2763	0.0816	0.1845	0.8104	0.3284	0.0736	0.1901	0.8224
t5-base	**0.4416**	**0.1361**	**0.1893**	**0.8288**	**0.4017**	**0.0928**	**0.2343**	**0.8419**

CHAPTER 5

SOCIAL IMPACT SUSTAINABILITY

Beyond its technical achievements, this multi-document summarization project is governed by a commitment to responsible innovation and sustainable development. This section evaluates the project's impact beyond its technical metrics, detailing its alignment with global sustainability principles and assessing its consequences on environmental, social, and governance (ESG) factors.

5.1 ALIGNMENT WITH GLOBAL SUSTAINABILITY GOALS

This project supports the goals outlined in the United Nations Sustainable Development Goals (SDGs), demonstrating a commitment to global sustainable development.

- **SDG 4 - Quality Education:** The core function of multi-document summarization enhances information accessibility by providing condensed knowledge, thereby supporting research and educational objectives for users processing large document sets, especially in the legal domain.
- **SDG 9 - Industry, Innovation, and Infrastructure:** This novel pipeline represents an advancement in robust information processing technology, contributing to efficient and reliable data generation within legal institutions.

- **SDG 16 - Peace, Justice, and Strong Institutions:** By enabling faster, unbiased, and comprehensive synthesis of policy, legal, or regulatory documents, the tool promotes transparency, accountability, and informed decision-making.

5.2 ENVIRONMENTAL, SOCIAL AND GOVERNANCE (ESG)

5.2.1 Environmental Sustainability

When implemented on a large scale, the system's implementation may lead to environmental concerns because of the collective energy used to infer and maintain the model over time. Both transformer models and Graph Neural Network (GNN) models consume a substantial amount of computational resources during training, which correspondingly, will generate significant energy consumption. The system's potential negative impact from energy networking can be reduced if the users take advantage of the System's ability to reuse pre-trained models, reduce retraining time, and optimize inference thresholds. Additionally, users who utilize digital summaries will have less reason to rely upon print, therefore reducing the overall environmental impact associated with both computation and lithographic printing.

5.2.2 Social Impact

The deployment of this system provides non-experts access to complex legal theories through the use of brief summaries. However, there is a chance that some complex legal concepts may be distilled down resulting in a partial or incomplete understanding of a legal argument. Tthe use of automated systems for text generation raises concerns regarding trust and dependence. Users may place too much trust in the resulting summary without checking for accuracy with the original source of information. To alleviate this concern, the system includes mechanisms for fact verification and has been intentionally created as a support system, not as a source of authority.

5.2.3 Governance

Future development is committed to transparent and ethical deployment practices. Governance standards for deployment will require continuous monitoring of summary quality, especially using metrics like ROUGE-L for structural completeness and human assessment of coherence, to uphold the quality and integrity of the informational output.

5.3 HEALTH CONSIDERATIONS

Regarding safety from a deployment perspective, reliability and correctness of generated summaries is the primary concern. The consequences of using incorrect or fictitious information to make a decision could have dire

consequences, with no way of verifying the information used to make the decision. As a means of preventing this risk, this system integrates supervised verification of facts through the use of Natural Language Inference (NLI) and maintains the chronological sequence of events through time-based references. Despite the inclusion of these safeguards, some level of risk still exists. Thus, human oversight is needed throughout the use of this system in the real world.

5.4 LEGAL CONSIDERATIONS

The implementation of the system within the legal field raises several issues from both a legal and ethical perspective regarding the way data is used, liability for any errors made by the system, and how data may be misinterpreted due to poor training of the underlying models. Even though the system relies entirely upon publicly available datasets such as MultiLexSum and Supreme Court of India for training, it is possible that users may mistakenly believe that the Generated summaries constitute ”Legal Advice.” To combat this possibility, the Project has developed the System as a decision support tool only and has taken steps to reduce the likelihood of users misinterpreting Judicial opinions. These include implementing fact verification to ensure that Judicial decisions are never misrepresented and to encourage use of the system for ethical and responsible uses.

CHAPTER 6

LIMITATIONS

The proposed framework does demonstrate strong performance, however it suffers multiple limitations. These include: reliance upon the accuracy of the underlying argument mining systems, sensitivity dependent upon the selection of hyperparametric thresholds, severely limited ability to generalize across disparate legal domains, the computational complexity of using graph based architectures, and reliance upon availability of evaluation metric tools that are purely based upon lexical measure i.e., ROUGE. Additionally, the system lacks end-to-end optimization, any external gleans of legal knowledge, and no expert human evaluators. Addressing these issues will be essential in addressing scalability, robustness, and compliance with real world use.

- **Dependence on Argument Mining Accuracy:** The total effectiveness of the entire framework is greatly dependent on how well the argument mining module identifies and classifies each sentence, such as Claim, Evidence, Rule, Background, etc. Errors in this classification will dramatically impact the result of the construction of the knowledge graph and the ranking of the sentences for the final output.

- **Sensitivity to Threshold Hyperparameters:** There are multiple threshold parameters in place for this framework when executing it properly. These include the threshold of similarity for constructing the knowledge graph, the threshold for filtering redundant documents, and the weighting factors used for ranking the various sentences. The performance of the framework

will vary considerably from one dataset to the next as well as across legal domains due to the sensitivity of these hyperparameters, therefore, each of these parameters needs to be manually tuned.

- **Limited Generalization Across Legal Domains:** The evaluation of the framework has primarily been done using the Multi-LexSum dataset in addition to a limited dataset of Indian legal documents. However, because Legal documents vary enormously depending on what field the law falls under, the way in which the system developed can produce generalizable results across different Legal domains will require some level of Domain specific adaptation and adjustment.

- **Computational Complexity and Scalability:** The employment of graphs based upon sentence-level similarity in conjunction with Graph Neural Networks (GCN/GAT) introduces significant overhead in terms of computational power and processing time. The process used to create these graphs requires the calculation of pairwise similarity for all sentences includes an additional cost due to the time needed for attention based message passing.

- **Limitations of Evaluation Metrics for Legal Reasoning:** Evaluation of the system relies primarily upon ROUGE and BERTScore. These are both measures of the similarity of lexical and semantic content, but do not directly correspond with preserving legal reasoning and the coherence of argumentative material nor checking for factual correctness, i.e., the system is constrained due to a lack of specialized evaluation metric tools and expert human validation.

CHAPTER 7

CONCLUSIONS AND FUTURE WORK

The objective of this study is to create a comprehensive legal summary system that can accurately summarize, process, and produce clear and concise summaries from the vast amounts of legal information available. The proposed system combines graph and transformer models to generate summaries from large bodies of law by providing relational data between individual pieces of information in the legal ecosystem such as case opinions, e.g., rule and precedent, and citation analysis to represent the interrelatedness of legal facts or legal concepts.

The primary challenges to the proposed system will be building a scalable way to represent and process complex legal documents and case collections within graph and transformer paradigms, developing algorithms that adapt to the continuous introduction of new legal case precedence, and gaining access to large amounts of annotated legal documents for further model training and development purposes.

Future work will consist of developing scalable graph creation and encoding processes for long legal documents and multiple case collections. This will involve: using Graph Convolutional Networks (GCNs), like GNNs, and Graph Attention Networks (GANs), to capture high-level, relational dependencies between legal cases, legal arguments, and legal citations, creating large, heterogeneous, and dynamic legal knowledge graphs that include concepts, events, and arguments, and then using GNN encoders and transformer-based decoders, in a multi-layer manner, to generate a summary of the legal information written by an entity.

To test these improvements, additional studies will need to be conducted over both benchmark datasets (like Multi-LexSum and BillSum), as well as corpus representations from more than one type of jurisdiction. One example could be Indian case law. Evaluation methods should not just consist of ROUGE/BERTScore but also include things such as factual positioning and temporal order of events. Parts of these models will be benchmarked by having ablation studies performed that vary other parameters like node degree (graph threshold) and create new graphs with different graphs by using various attention mechanisms (T5/BART versus other available open-source LLMs e.g., Mistral/LLaMA).

By integrating the two reasoning methods/models (GNN-based reasoning and fact verification modules), factual accuracy is expected to be improved (less hallucinations) and legal argument flow preserved. Also, the use of a retrieval-augmented verification pipeline will substantially improve the overall reliability and explanatory capability of claims generated through documented sources.

In conclusion, combining graph-based reasoning and retrieving-augmented verification with large language models will create a verifiable, explainable system of summarization that supports real-world legal decisions.

In the future, emerging and open-source LLMs (for e.g., Mistral and Llama) and pre-training frameworks for document summary generation can represent further advancements in legal summarization. Integrating graph-based reasoning and annotated information retrieval, in addition to claim verification, will produce explainable, verifiable, and legally reliable AI-based summary generation systems.

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